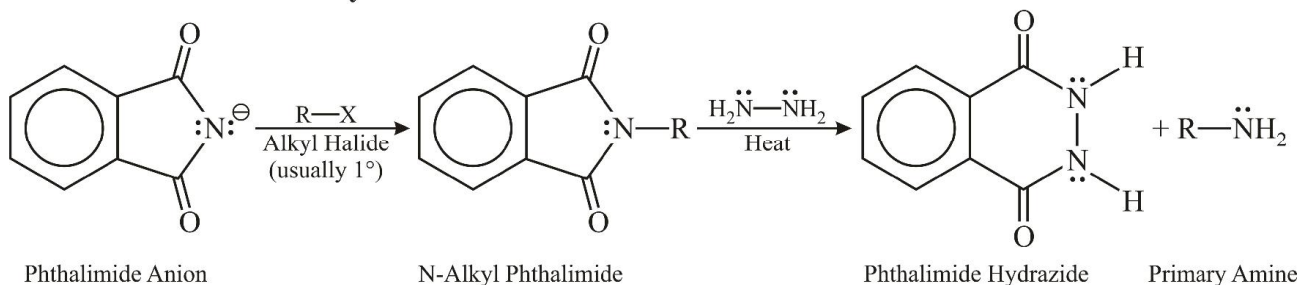


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AMINES

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Gabriel's Phthalimide Synthesis



THEORY

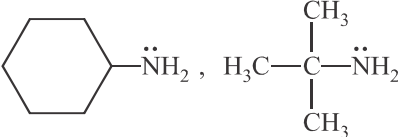
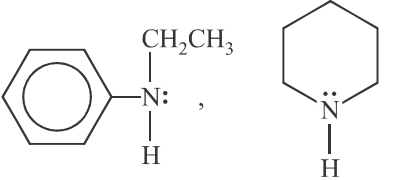
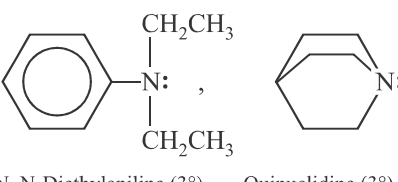
1. INTRODUCTION

Amines are organic derivatives of ammonia with one or more alkyl or aryl groups bonded to the nitrogen atom. Amines serve many functions in living organisms, such as bioregulation, neurotransmission, and defense against predators. Because of their high degree of biological activity, many amines are used as drugs and medicines.

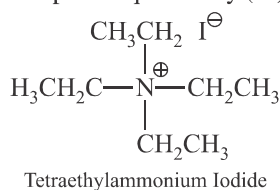
The alkaloids are an important group of biologically active amines, mostly synthesized by plants to protect them from being eaten by insects and other animals. Although some alkaloids are used medicinally (chiefly as painkillers), all alkaloids are toxic and cause death if taken in large quantities.

2. CLASSIFICATION

Amines are classified as **primary** (1°), **secondary** (2°), or **tertiary** (3°), corresponding to one, two or three alkyl or aryl groups bonded to nitrogen.

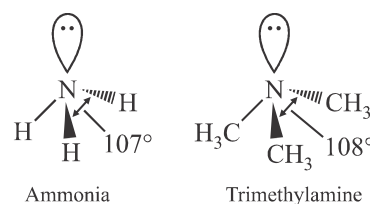
Type	Examples
Primary (1°) $R-NH_2$	 Cyclohexylamine (1°) Tert-Butylamine (1°)
Secondary (2°) R_2NH	 N-Ethylaniline (2°) Piperidine (2°)
Tertiary (3°) R_3N	 N, N-Diethylaniline (3°) Quinuclidine (3°)

Quaternary ammonium salts have four alkyl or aryl bonds to a nitrogen atom. The nitrogen atom bears a positive charge, just as it does in simple ammonium salts such as ammonium chloride. The following are examples of quaternary (4°) ammonium salts.

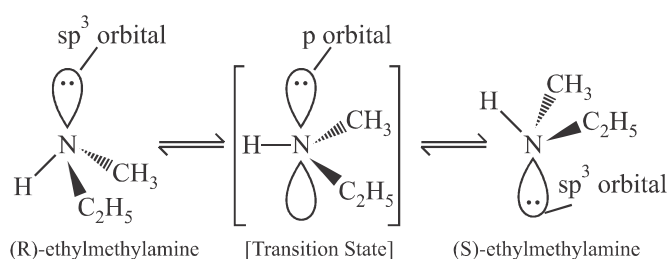


3. STRUCTURE OF AMINES

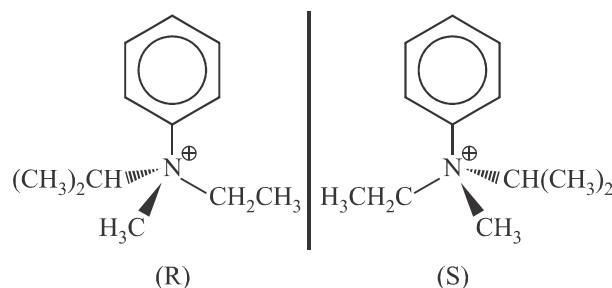
Ammonia has a slightly distorted tetrahedral shape. A lone pair of nonbonding electrons occupies one of the tetrahedral positions. This geometry is represented by sp^3 hybridization of nitrogen, with the bulky lone pair compressing the $H-N-H$ bond angles to 107° from the "ideal" sp^3 bond angle of 109.5° . Trimethylamine shows less angle compression because the bulky methyl groups open the angle slightly.



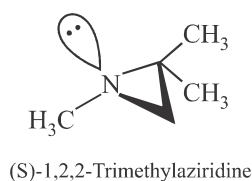
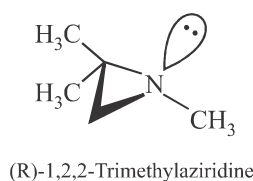
A tetrahedral amine with three different substituents (and a lone pair) is non-superimposable on its mirror image. In most cases, we cannot resolve such an amine into two enantiomers, however, because the enantiomers interconvert rapidly. This interconversion takes place by **nitrogen inversion**, in which the lone pair moves from one face of the molecule to the other.



In case of quaternary ammonium salts with asymmetric nitrogen atoms, inversion of configuration is not possible because there is no lone pair to undergo nitrogen inversion.



Amines that cannot attain the sp^2 -hybrid transition state for nitrogen inversion also show chirality. If the nitrogen atom is contained in a small ring, for example, it is prevented from attaining the 120° bond angles that facilitate inversion



4. PHYSICAL PROPERTIES

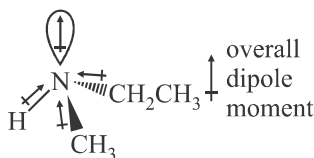
4.1 State

The lower aliphatic amines are gases with fishy odour. Primary amines with three or more carbon atoms are liquid and higher ones are solid.

Aniline and other aryl amines are usually colourless but gets coloured on storage due to atmospheric oxidation.

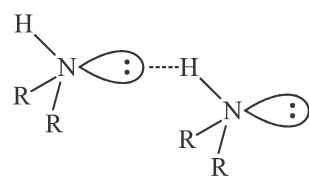
4.2 Dipole Moment

Amines are strongly polar because the large dipole moment of the lone pair of electrons adds to the dipole moments of the $C \rightarrow H$ and $H \rightarrow N$ bonds.

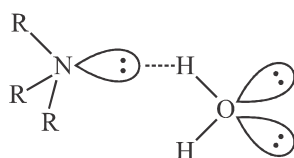


4.3 Solubility

Primary and secondary amines have $N-H$ bonds, allowing them to form hydrogen bonds. Having no $N-H$ bonds, pure tertiary amines cannot engage in hydrogen bonding. They can, however, accept hydrogen bonds from molecules having $O-H$ or $N-H$ bonds.



1° or 2° Amine:
Hydrogen Bond Donor and Acceptor



3° Amine:
Hydrogen Bond Acceptor only

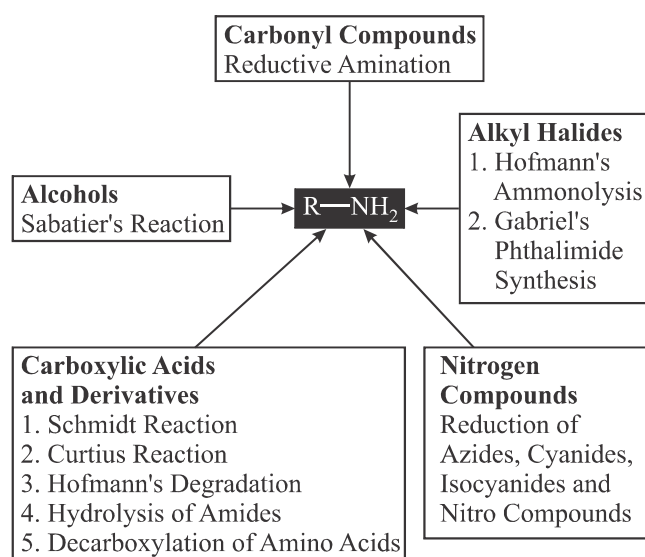
Therefore, lower aliphatic amines are soluble in water because they can form hydrogen bonds with water. However, solubility decreases with increase in molar mass of amines due to increase in size of the hydrophobic alkyl part. Higher amines are essentially insoluble in water.

4.4 Boiling Point

Because nitrogen is less electronegative than oxygen, the $N-H$ bond is less polar than $O-H$ bond. Therefore, amines form weaker hydrogen bonds than do alcohols of similar molecular weights. Primary and secondary amines have boiling points that are lower than those of alcohols, yet higher than those of ethers of similar molecular weights. With no hydrogen bonding, tertiary amines have lower boiling points than primary and secondary amines of similar molecular weights.

Compound	BP(°C)	Type	Molecular Weight
$(CH_3)_3N$:	3	Tertiary Amine	59
$CH_3-O-CH_2-CH_3$	8	Ether	60
$CH_3-NH-CH_2-CH_3$	37	Secondary Amine	59
$CH_3CH_2CH_2-NH_2$	48	Primary Amine	59
$CH_3CH_2CH_2-OH$	97	Alcohol	60

5. PREPARATION OF AMINES



5.1 Alkyl Halides

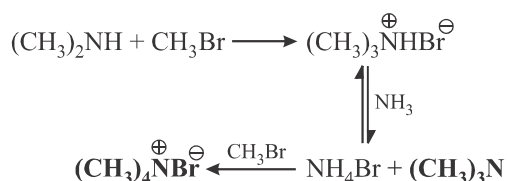
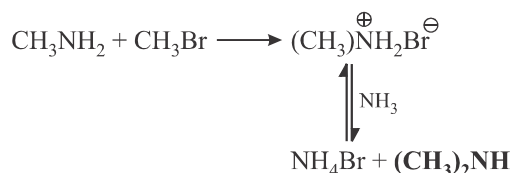
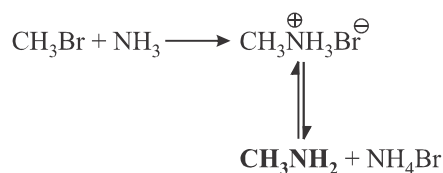
Example - 2

5.1.1 Hofmann's Ammonolysis Method

It is an **ammonolysis**. An alkyl halide and an ethanolic solution of ammonia are heated in a sealed tube at 100°C.

- The process gives a mixture of products.
- Tertiary alkyl halide is not suitable as it gives an alkene.
- Reaction follows bimolecular substitution (S_N2).

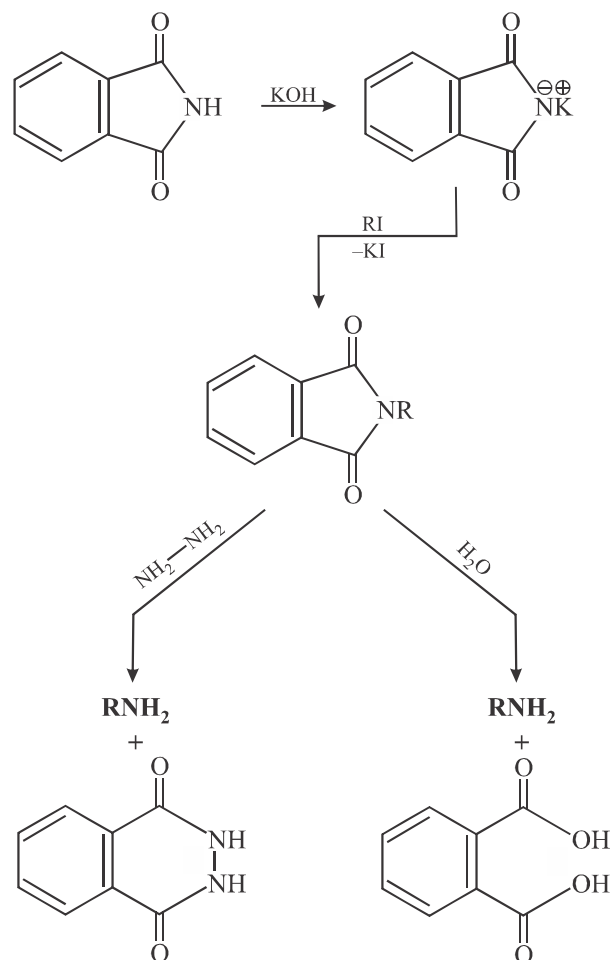
Example - 1



5.1.2 Gabriel's Phthalimide Synthesis

Phthalamide is treated with KOH first and then potassium phthalamide is heated with alkyl halide to yield alkyl phthalamide. The subsequent hydrolysis produces exclusively 1° amine.

Alkyl phthalamide is also treated with hydrazine to get a better yield of 1° amine.



5.1.3 Grignard Reagent

Grignard reagent or a trialkyl borane on treatment with chloramine gives the primary amine.

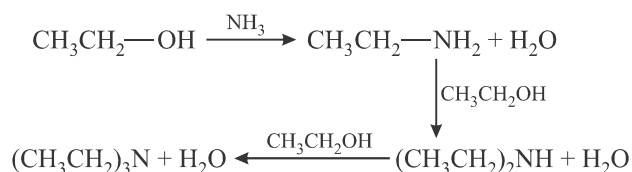


5.2 Alcohols

5.2.1 Sabatier Reaction

Alcohols and ammonia are heated under pressure in the presence of a catalyst e.g., **copper chromite or alumina**. A mixture of products is obtained.

Example - 3



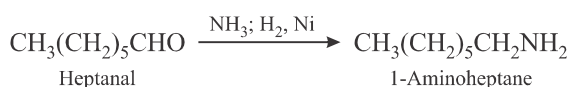
5.3 Carbonyl Compounds

5.3.1 Reductive Amination

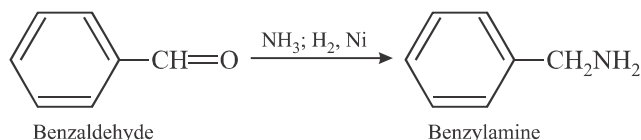
Several aldehydes and ketones are converted into amines by reduction in the presence of ammonia.

Reduction can be accomplished catalytically, (**hydrogen (20-150 atm) over Raney nickel at 40-150°C** or by use of **sodium cyanohydridoborate, NaBH₃CN**).

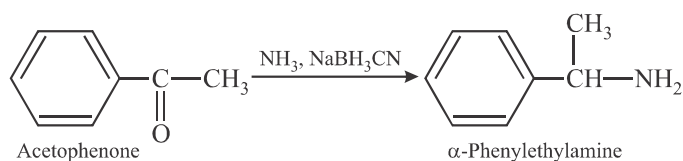
Example - 4



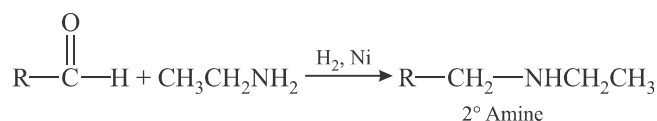
Example - 5



Example - 6



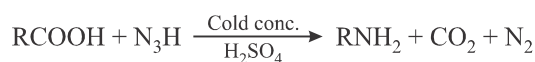
Example - 7



5.4 Carboxylic Acids & Derivatives

5.4.1 Schmidt Reaction

A treatment of cold conc. H₂SO₄ to the mixture of carboxylic acid and hydrazoic acid gives 1° amine.

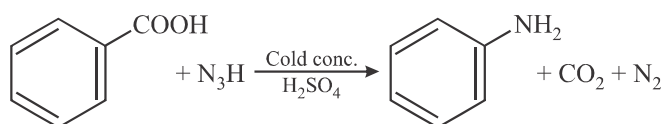


Note...

1. The product molecule contains one carbon atom less than that in acid.

2. Isocyanate is formed as an intermediate in the reaction.

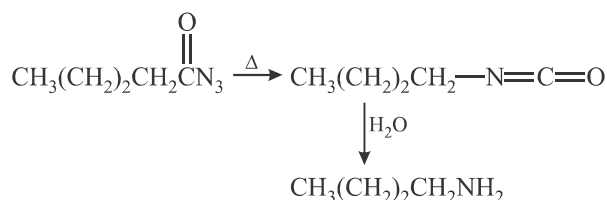
Example - 8



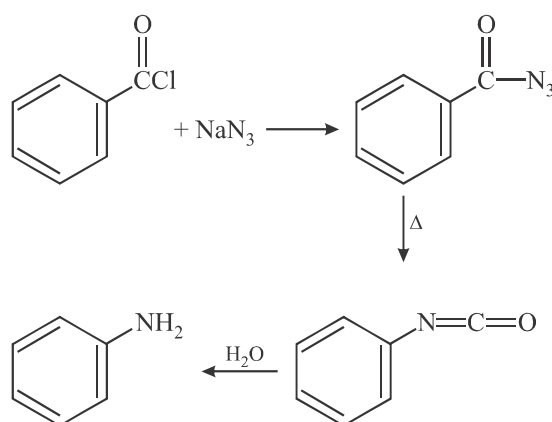
5.4.2 Curtius Reaction

The reaction involves the pyrolysis of acyl azide to produce isocyanates. The subsequent hydrolysis gives amine.

Example - 9



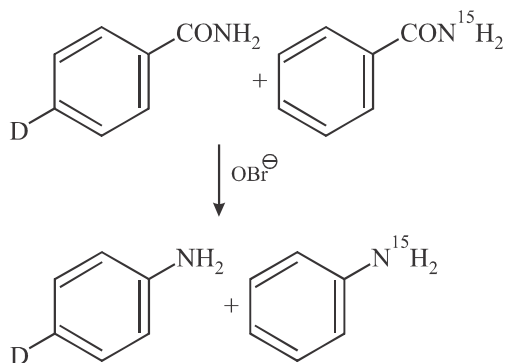
Example - 10



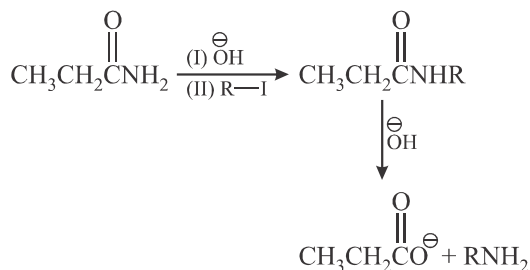
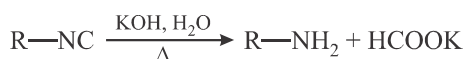
5.4.3 Hofmann's Degradation of Amides

Amide is warmed with bromine and concentrated aqueous KOH or NaOH solution.

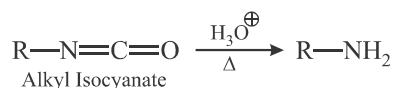
- (a) Product is 1° amine having one carbon less than amide.
 (b) Alkyl isocyanate is the intermediate and rearrangement is intramolecular.

Example - 11**5.4.4 Hydrolysis of Amides & Isocyanides**

N-substituted amide and isocyanide on hydrolysis give 1° amine.

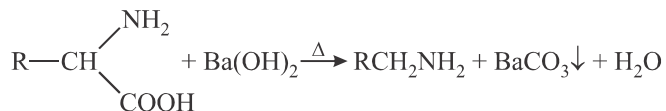
Example - 12**Example - 13**

Hydrolysis of Isocyanates also yields amines.

**5.4.5 Decarboxylation of Amino Acids**

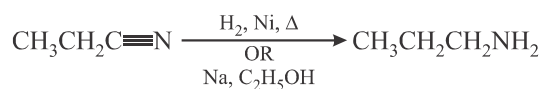
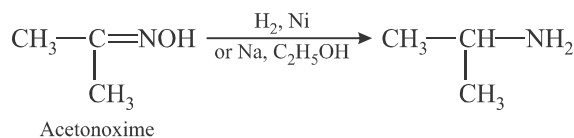
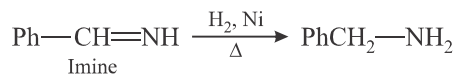
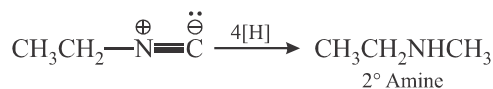
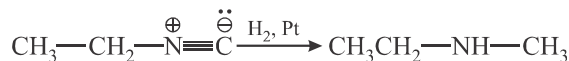
Amino acids on heating with barium hydroxide gives 1° amine.

Quaternary ammonium hydroxide on heating gets decomposed into alcohol if β-hydrogen is not present in the alkyl group but if there is β-hydrogen, **Hofmann elimination** proceeds.

**5.5 Nitrogen Compounds****5.5.1 Reduction of Cyanides, Isocyanides, Oximes, Imines**

Nitrile (—CN), oxime (=NOH), imine (=NH), enamine etc. on catalytic reduction with H₂, Ni give the corresponding amine.

- Note...*
1. The reduction of nitrile by Na/C₂H₅OH is known as **Mendius reaction**.
 2. In most of the cases LiAlH₄ is also effective for reduction.
 3. Reduction of alkyl isocyanides with Na/C₂H₅OH or H₂/Pt gives 2° amine.

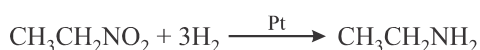
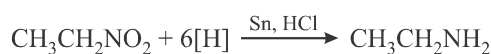
Example - 14**Example - 15****Example - 16****Example - 17****Example - 18****5.5.2 Reduction of Nitro Compounds**

Nitroalkanes are usually reduced by tin and hydrochloric acid or lithium aluminium hydride into corresponding amine. H₂ and catalyst is also used for such a reduction.

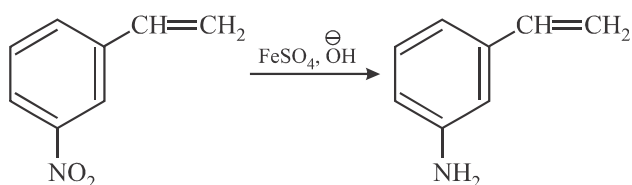
Note.. 1. 1° aromatic amines are usually prepared by the reduction of corresponding nitro compounds by **tin, iron or zinc with HCl or CH₃COOH**.

2. Aqueous ethanolic ammonium hydrogen sulphide, aqueous sodium sulphide or methanoic sodium hydrogen sulphide selectively reduce one nitro group into a polynitro compound.

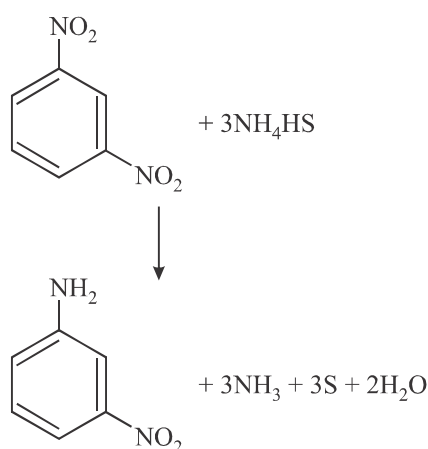
Example - 19



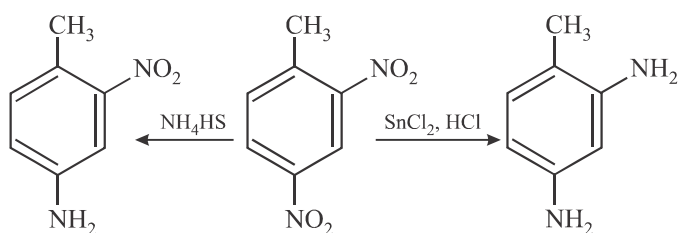
Example - 20



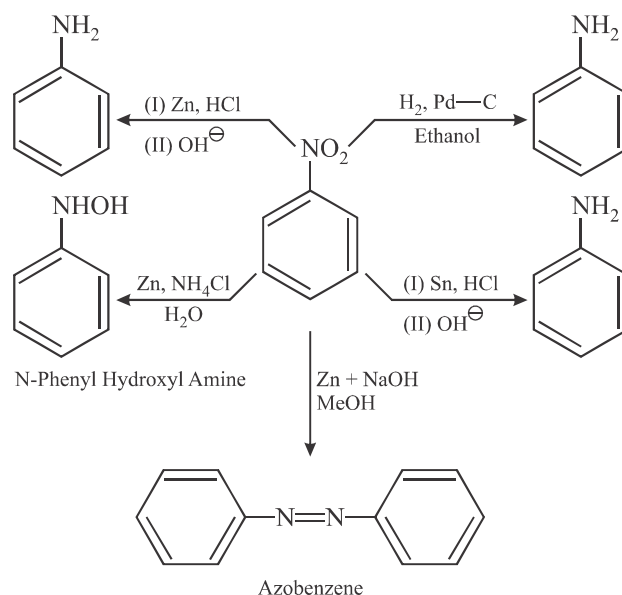
Example - 21



Example - 22



Example - 23

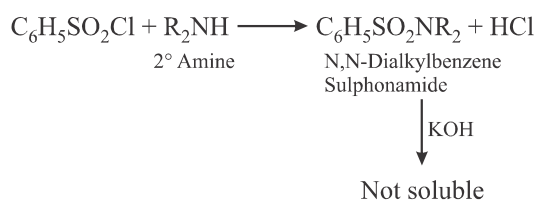
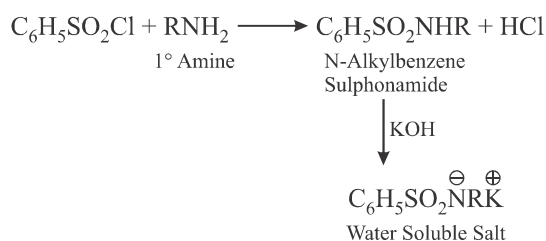


6. REACTIONS OF AMINES

6.1 Sulphonylation – Hinsberg Test

Treatment with benzene sulphonyl chloride (**Hinsberg reagent**) or p-toluene sulphonyl chloride.

- The reaction is used to separate the amine mixture.
- 1° and 2° amine, due to the presence of active hydrogen react and give corresponding sulphonamide while 3° amine does not react.
- 1° amine product N-alkyl benzene sulphonamide is soluble in KOH forming a water-soluble salt.



6.2 Nitrous Acid Test

- (a) 1° amine on reaction with HNO_2 gives usually an alcohol. The reaction involves the formation of diazonium salt as an intermediate.
- (b) 2° amine gives nitrosoamine which is a yellow oily liquid.
- (c) 3° amine dissolves in cold HNO_2 to form an unstable salt which on heating decomposes to give nitrosoamine and an alcohol.

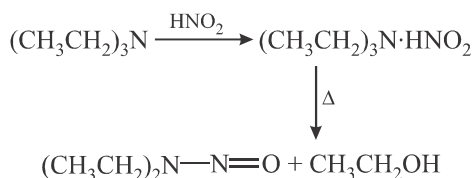
Example - 24



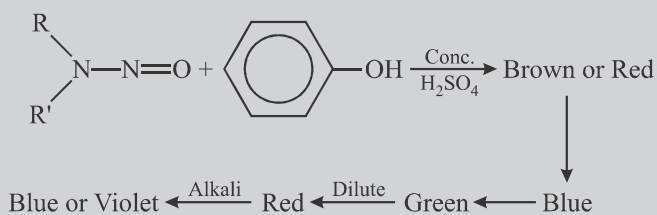
Example - 25



Example - 26



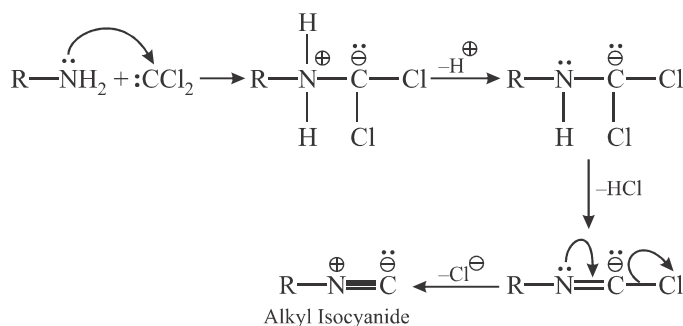
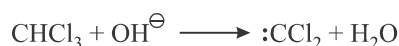
Note... **Liebermann's Nitroso Test** - Test for 2° amines (nitroso amines) and phenols.



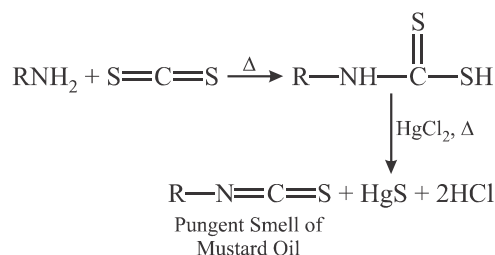
6.3 Carbylamine Reaction

1° amine on treatment with chloroform and alcoholic KOH gives out unpleasant or pungent vapours of alkyl carbylamine (isocyanide).

The reaction is known as a test for 1° amine.

6.4 Treatment with $\text{CS}_2/\text{HgCl}_2$

- (a) 1° amine on warming with CS_2 produces dithiocarbamic acid, which is decomposed by mercuric chloride to alkyl isothiocyanate. It is called as **Hofmann's Mustard Oil Reaction**.
- (b) Although 2° amine gives dithiocarbamic acid but is not decomposed by mercuric chloride.
- (c) 3° amine don't react with CS_2 .



6.5 Acylation

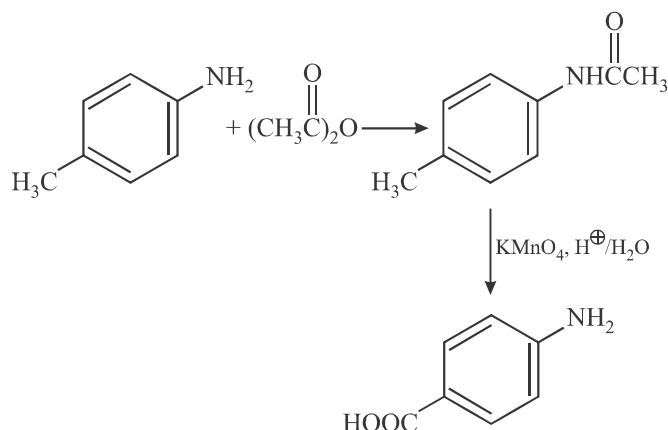
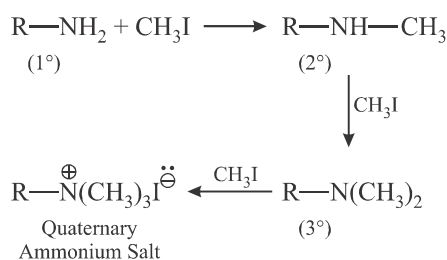
1° and 2° amines react with acid halide or anhydride to give N-substituted amides.

For aniline, the reaction is used to protect the ring.

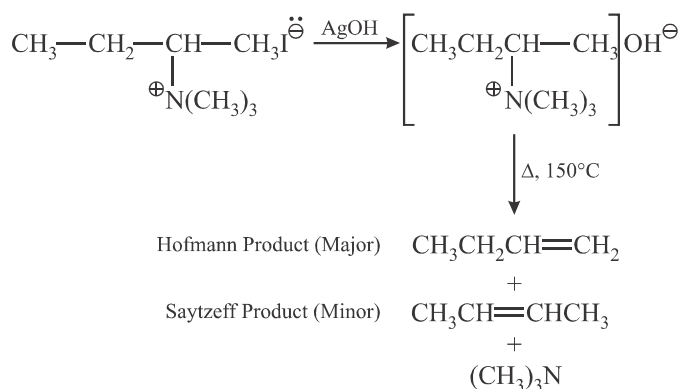
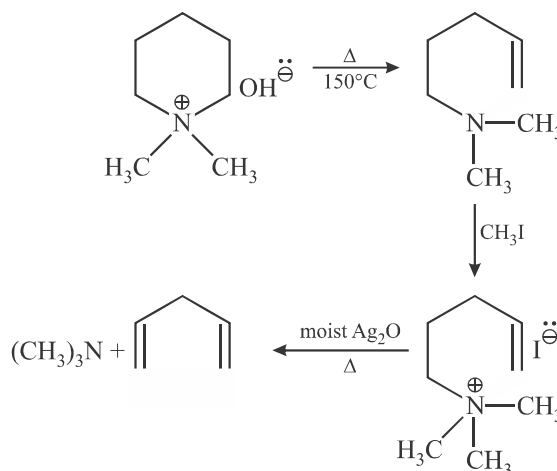
Example - 27



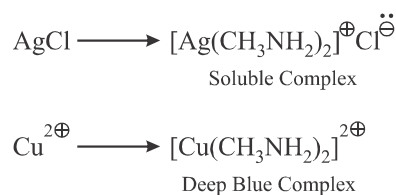
Example - 28

Example - 29

6.6 Hofmann's Exhaustive Methylation and Elimination


Quaternary Ammonium Iodides are converted to hydroxides with moist Ag_2O . The hydroxide on heating undergo elimination to give 3° amines and alkenes. The important aspect of this elimination is that the least substituted alkene is the major product (**Hofmann Rule**). This is known as **Hofmann's Elimination**.

Example - 30

Example - 31

6.7 Metal Ions

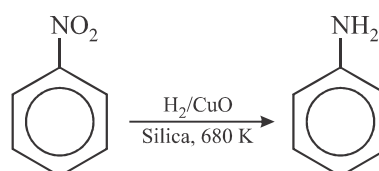
Amines form coordination compounds with metal ions.

Example - 32

7. PREPARATION OF AROMATIC AMINES

Besides the methods in the preparation of aliphatic amines, following methods can also be used.

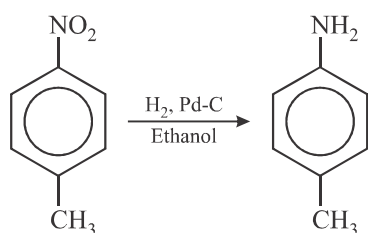
7.1 Reduction of Nitro Compounds
7.1.1 Vapour-Phase Reduction

Reducing Agents : CuO on SiO_2 or V-Pt Catalyst.

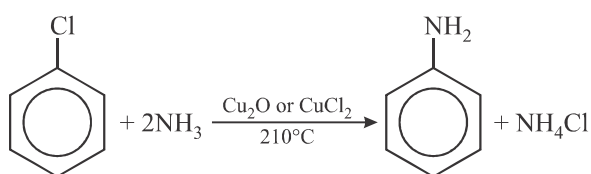
Example - 33


7.1.2 Catalytic Hydrogenation

Reducing Agents : H_2 , Pd-C/Et-OH.

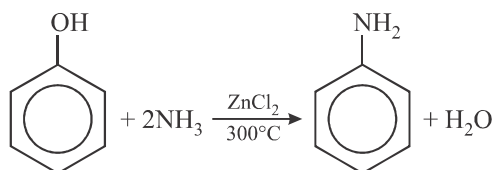


7.2 Ammonolysis of Aryl Halides

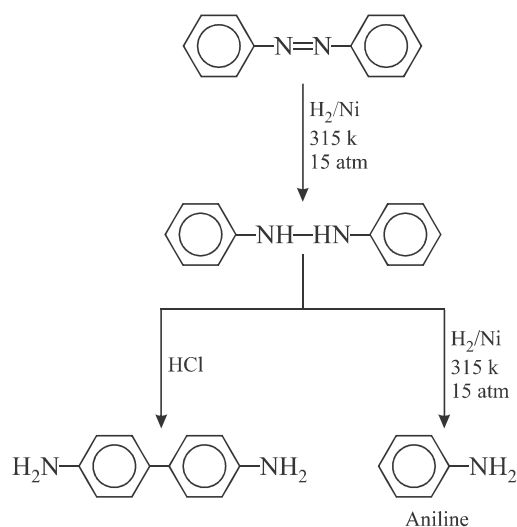


Note... If deactivating groups ($-\text{NO}_2$ etc.) are at *ortho* or *para* position, reaction becomes quite easy to carry out.

7.3 Phenol and Ammonia

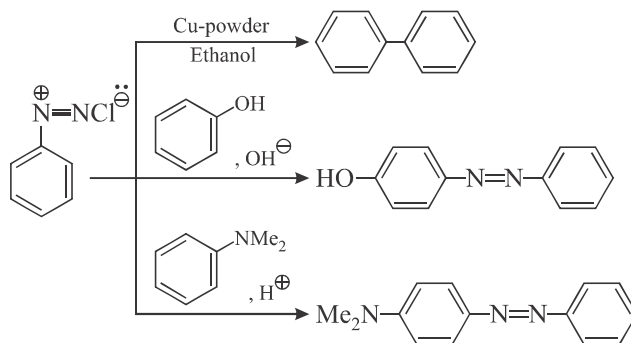
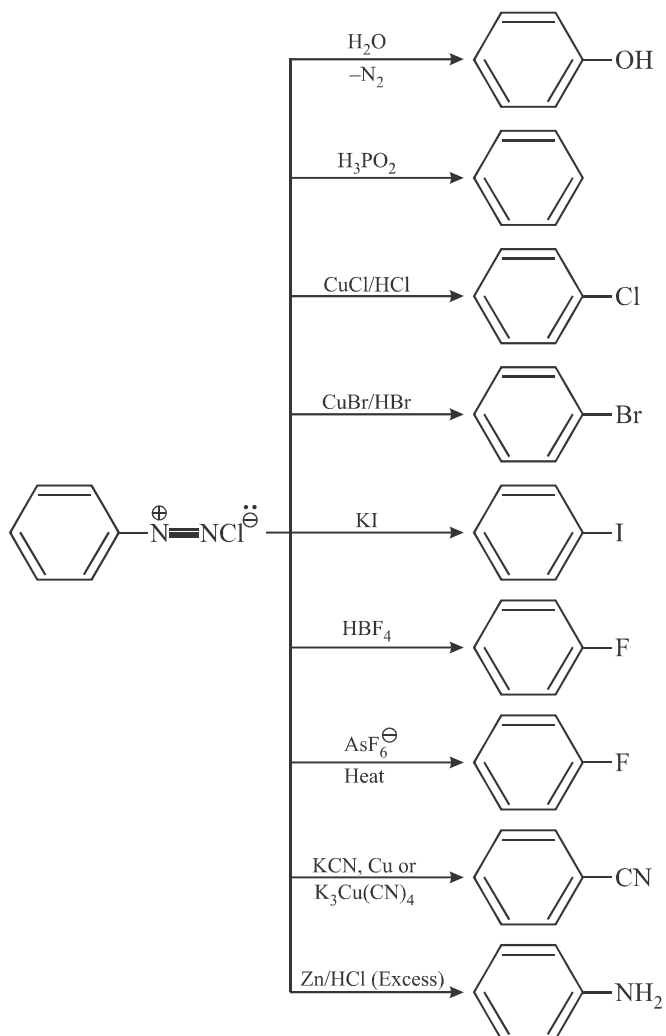
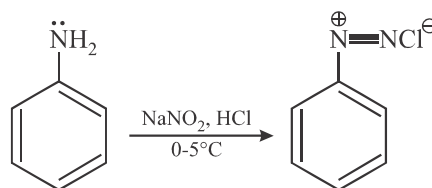


7.4 Reduction of Azo Compounds



8. REACTIONS OF AROMATIC AMINES

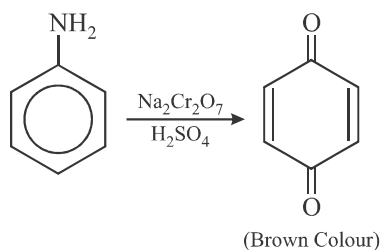
8.1 Diazonium Salt and Reactions



8.2 Oxidation

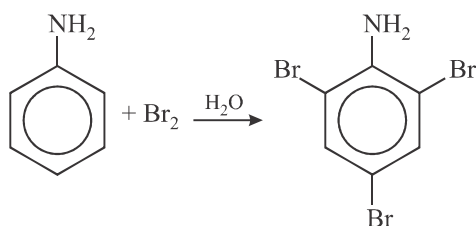
Aromatic amines are readily oxidised to various products depending on conditions. Under controlled oxidation using $\text{Na}_2\text{Cr}_2\text{O}_7$ and H_2SO_4 , aniline is oxidised to p-benzoquinone. In fact, on standing in air, p-benzoquinone is formed.

If oxidation is not controlled, a black dye, aniline black is formed.



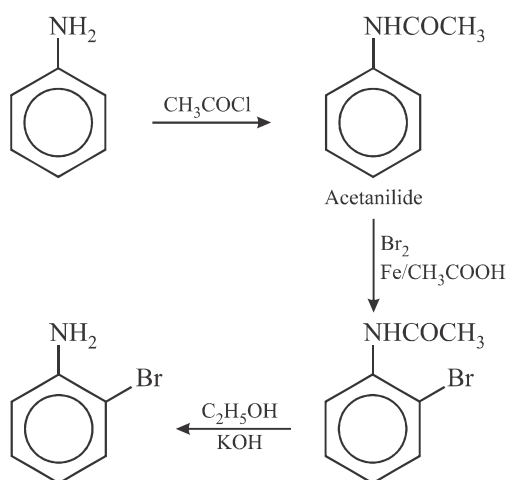
8.3 Halogenation

Amine group (NH_2) is a very strong activating group and an ortho/para director. It forms 2, 4, 6-tribromo aniline on normal bromination in H_2O .



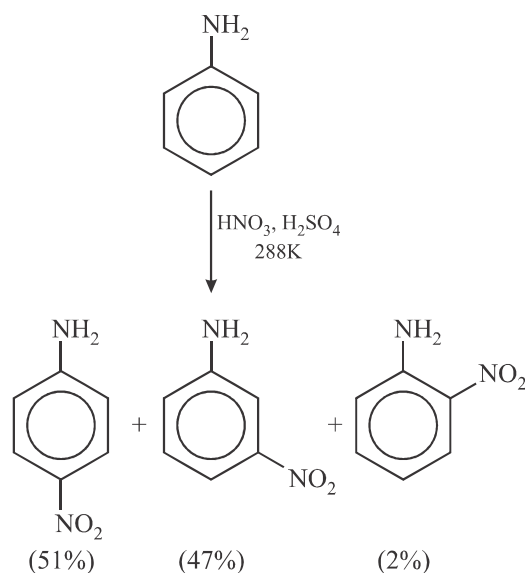
ortho and para derivatives can be made by decreasing the activating power of amino group by acetylation of aniline. $-\text{NHCOCH}_3$ group is a mild ortho and para director.

Example - 34

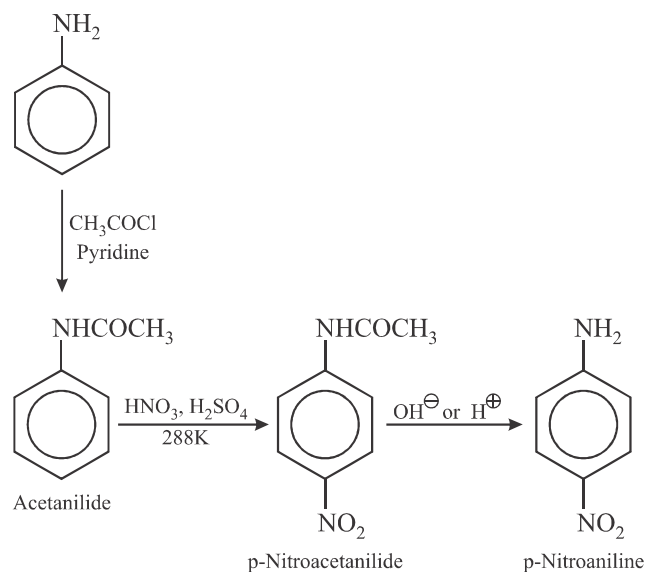


8.4 Nitration

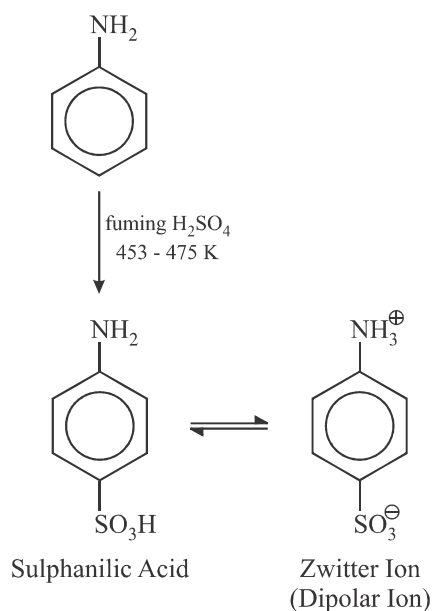
The normal nitration of aniline is not done as nitrating mixture is oxidising and NH_2 is a strong activating group. Moreover, at 288 K aniline is protonated to form the anilinium ion which is meta directing. That is why besides the ortho and para derivatives, a significant amount of meta derivative is also formed.



However, by protecting the $-\text{NH}_2$ group by acetylation reaction with acetic anhydride, the nitration reaction can be controlled and the p-nitro derivative can be obtained as the major product.



8.5 Sulphonation



Note... Usually $-\text{NH}_2$ group attached to benzene is unable to form zwitter ion but here $-\text{SO}_3\text{H}$ is very strongly acidic. Hence sulphanilic acid forms zwitter ion. For example when $-\text{NH}_2$ group is attached to benzoic acid in any position, zwitter ion is not formed.

9. ANALYSIS OF AMINES

Amines are characterized chiefly through their basicity. A water-insoluble compound that dissolves in cold dilute hydrochloric acid or a water-soluble compound whose aqueous solution turns litmus blue - must almost certainly be an amine.

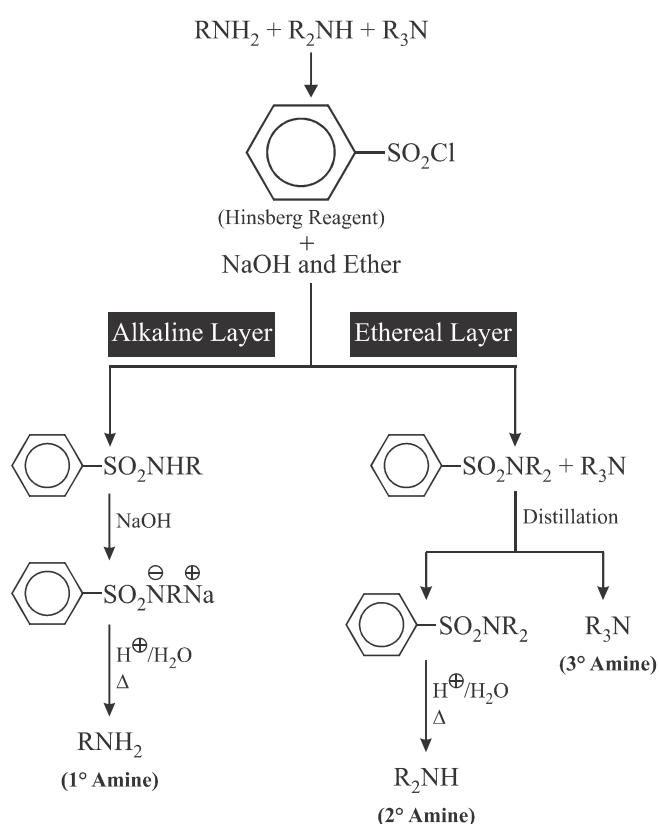
Whether an amine is primary, secondary, or tertiary is best shown by the **Hinsberg test**. The amine is shaken with benzenesulfonyl chloride in the presence of aqueous potassium hydroxide. Primary and secondary amines form substituted sulfonamides; tertiary amines do not.

The monosubstituted sulfonamide from a primary amine has an acidic hydrogen attached to nitrogen. Reaction with potassium hydroxide converts this amide into a soluble salt.

The disubstituted sulfonamide from a secondary amine has no acidic hydrogen and remains insoluble in the alkaline reaction mixture.

Behaviour toward nitrous acid is of some use in determining the class of an amine. In particular, the behaviour of primary aromatic amines is quite characteristic: treatment with nitrous acid converts them into diazonium salts, which yield highly colored azo compounds upon treatment with β -naphthol.

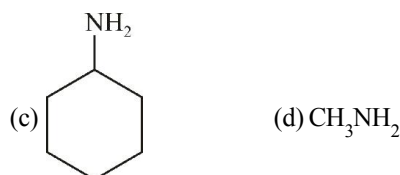
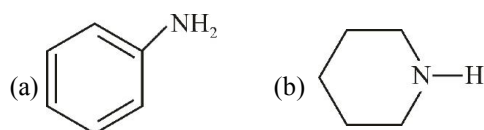
Hinsberg test is also used for the separation of amines.



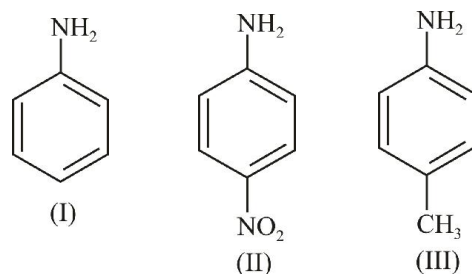
EXERCISE - 1 : BASIC OBJECTIVE QUESTIONS

Properties and Nomenclature

1. Which of the following is a 3° amine ?
(a) 1-methylcyclohexylamine
(b) Triethylamine
(c) tert-butylamine (d) N-methylaniline
2. The correct IUPAC name for $\text{CH}_2=\text{CHCH}_2\text{NHCH}_3$ is
(a) Allylmethylamine (b) 2-amino-4-pentene
(c) 4-aminopen-1-ene
(d) N-methylprop-2-en-1-amine
3. Amongst the following, the strongest base in aqueous medium is
(a) CH_3NH_2 (b) NCCH_2NH_2
(c) $(\text{CH}_3)_2\text{NH}$ (d) $\text{C}_6\text{H}_5\text{NHCH}_3$
4. Which of the following is the weakest Bronsted base ?

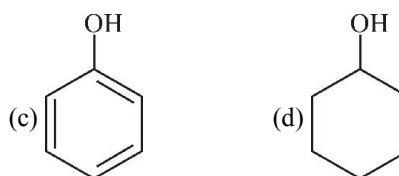
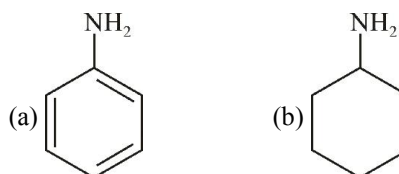


5. The correct increasing order of basic strength for the following compounds is

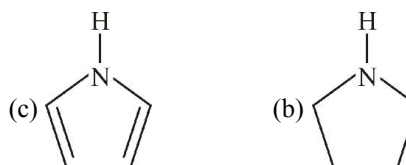
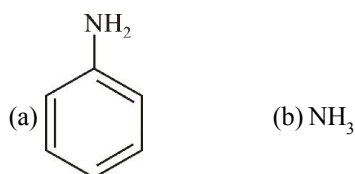


- (a) $\text{II} < \text{III} < \text{I}$ (b) $\text{III} < \text{I} < \text{II}$
(c) $\text{III} < \text{II} < \text{I}$ (d) $\text{II} < \text{I} < \text{III}$

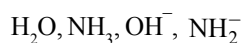
6. Which of the following compounds is the weakest Bronsted base ?



7. Among the following amines, the strongest Bronsted base is

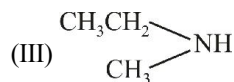
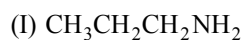


8. The correct decreasing order of basic strength of the following species is



- (a) $\text{NH}_2^- > \text{OH}^- > \text{NH}_3 > \text{H}_2\text{O}$
(b) $\text{OH}^- > \text{NH}_2^- > \text{H}_2\text{O} > \text{NH}_3$
(c) $\text{NH}_3 > \text{H}_2\text{O} > \text{NH}_2^- > \text{OH}^-$
(d) $\text{H}_2\text{O} > \text{NH}_3 > \text{OH}^- > \text{NH}_2^-$

9. Which of the following should be most volatile ?



(a) II (b) IV

(c) I (d) III

10. Which of the following is a secondary amine ?

(a) 2-Butanamine (b) N-Methylpiperidine

(c) N-Methyl-2-pentanamine

(d) p-Anisidine

11. Arrange the following in increasing order of their basic strength :



(a) $\text{IV} < \text{III} < \text{II} < \text{I}$ (b) $\text{IV} < \text{III} < \text{I} < \text{II}$

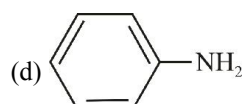
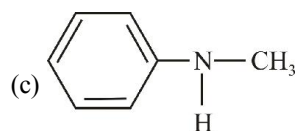
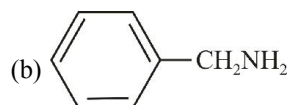
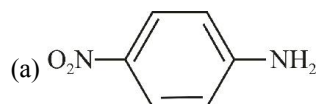
(c) $\text{I} < \text{II} < \text{III} < \text{IV}$ (d) $\text{IV} < \text{III} < \text{I} = \text{II}$

12. The weakest base among the following is

(a) dimethylamine (b) aniline

(c) methylamine (d) ethylamine

13. Which of the following compounds is most basic ?



14. The strongest base in aqueous solution among the following amines is

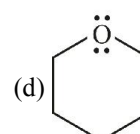
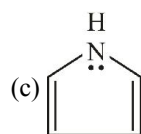
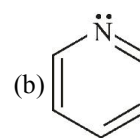
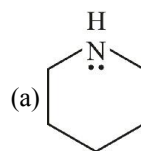
(a) N, N-diethylethanamine

(b) N-ethylethanamine

(c) N-methylmethanamine

(d) ethanamine

15. Strongest base is



16. $\text{C}_3\text{H}_9\text{N}$ represents

(a) Primary amine

(b) Secondary amine

(c) Tertiary amine

(d) All of these

17. A secondary amine is

(a) An organic compound with two $-\text{NH}_2$ groups

(b) A compound with two carbon atoms and an $-\text{NH}_2$ group

(c) A compound with an $-\text{NH}_2$ group on the carbon atom in number 2 position

(d) A compound in which two of the hydrogens of NH_3 have been replaced by organic groups

18. Allyl isocyanide has

(a) 9 sigma bonds and 4 pi bonds

(b) 8 sigma bonds and 5 pi bonds

(c) 8 sigma bonds, 3 pi bonds and 4 non-bonding electrons

(d) 9 sigma bonds, 3 pi bonds and 2 non-bonding electrons

19. $\text{CH}_2=\text{CH}-\text{CH}_2-\text{NH}-\text{CH}_3$ is a

(a) Secondary amine

(b) Primary amine

(c) Tertiary amine

(d) None of these

20. Leakage of which gas was responsible for the Bhopal tragedy in 1984

(a) $\text{CH}_3-\text{N}=\text{C}=\text{O}$

(b) $\text{CH}_3-\text{N}=\text{C}=\text{S}$

(c) CHCl_3

(d) $\text{C}_6\text{H}_5\text{COCl}$

21. How many primary amines are possible for the formula $C_4H_{11}N$?
- (a) 1 (b) 2
(c) 3 (d) 4

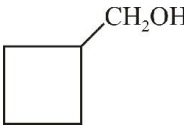
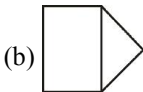
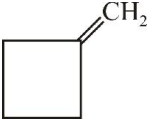
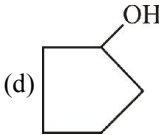
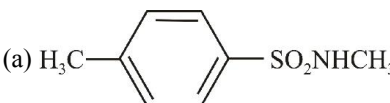
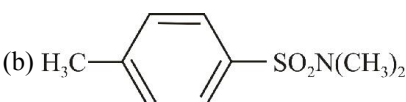
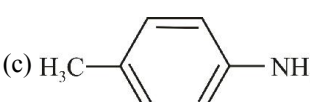
Preparation

22. In order to prepare a 1° amine from an alkyl halide with simultaneous addition of one CH_2 group in the carbon chain, the reagent used as source of nitrogen is
- (a) Sodium amide, $NaNH_2$
(b) Sodium azide, NaN_3
(c) Potassium cyanide, KCN
(d) Potassium phthalimide, $C_6H_4(CO)_2N^-K^+$
23. The source of nitrogen in Gabriel synthesis of amines is
- (a) Sodium azide, NaN_3 (b) Sodium nitrite, $NaNO_2$
(c) Potassium cyanide, KCN
(d) Potassium phthalimide, $C_6H_4(CO)_2N^-K^+$
24. Amongst the given set of reactants, the most appropriate for preparing 2° amine is
- (a) $2^\circ R-Br + NH_3$
(b) $2^\circ R-Br + NaCN$ followed by H_2/Pt
(c) $1^\circ R-NH_2 + RCHO$ followed by H_2/Pt
(d) $1^\circ R-Br$ (2 mol) + potassium phthalimide followed by $H_3O^+/heat$
25. The best reagent for converting 2-phenylpropanamide into 2-phenylpropanamine is
- (a) excess H_2 (b) Br_2 in aqueous NaOH
(c) iodine in the presence of red phosphorus
(d) $LiAlH_4$ in ether
26. The best reagent for converting, 2-phenyl-propanamide into 1-phenylethanamine is
- (a) excess H_2/Pt (b) NaOH/ Br_2
(c) $NaBH_4$ /methanol (d) $LiAlH_4$ /ether
27. Hofmann Bromamide Degradation reaction is shown by
- (a) $ArNH_2$ (b) $ArCONH_2$
(c) $ArNO_2$ (d) $ArCH_2NH_2$

28. Best method for preparing primary amines from alkyl halides without changing the number of carbon atoms in the chain is
- (a) Hofmann Bromamide reaction
(b) Gabriel phthalimide synthesis
(c) Sandmeyer reaction
(d) Reaction with NH_3
29. Which of the following methods of preparation of amines will not give same number of carbon atoms in the chain of amines as in the reactant ?
- (a) Reaction of alkyl nitrile with $LiAlH_4$.
(b) Reaction of amide with $LiAlH_4$ followed by treatment with water.
(c) Heating alkyl halide with potassium salt of phthalimide followed by hydrolysis.
(d) Treatment of amide with bromine in aqueous solution of sodium hydroxide.
30. Mendius reduction converts an alkyl cyanide to
- (a) a primary amine (b) an aldehyde
(c) a ketone (d) an oxime

Chemical Reactions

31. Benzylamine may be alkylated as shown in the following equation :
- $$C_6H_5CH_2NH_2 + R-X \longrightarrow C_6H_5CH_2NHR$$
- Which of the following alkyl halides is best suited for this reaction through S_N1 mechanism ?
- (a) CH_3Br (b) C_2H_5Br
(c) $C_6H_5CH_2Br$ (d) None of these
32. Methylamine reacts with HNO_2 to form
- (a) $CH_3-O-N=O$ (b) CH_3-O-CH_3
(c) CH_3OH (d) CH_3CHO
33. The gas evolved when methylamine reacts with nitrous acid is
- (a) NH_3 (b) N_2
(c) H_2 (d) C_2H_6
34. Acid anhydrides on reaction with primary amines give
- (a) amide (b) imide
(c) secondary amine (d) imine

35. An organic compound 'A' on treatment with NH_3 gives 'B' which on heating gives 'C', 'C' when treated with Br_2 in the presence of KOH produces ethylamine. Compound 'A' is
- (a) CH_3COOH (b) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$
- (c) $\text{CH}_3-\underset{\text{CH}_3}{\underset{|}{\text{CH}}}\text{COOH}$ (d) $\text{CH}_3\text{CH}_2\text{COOH}$
36. In the chemical reaction,
- $$\text{CH}_3\text{CH}_2\text{NH}_2 + \text{CHCl}_3 + 3 \text{KOH} \longrightarrow (\text{A}) + (\text{B}) + 3\text{H}_2\text{O}$$
- the compound (A) and (B) are respectively.
- (a) $\text{C}_2\text{H}_5\text{NC}$ and 3 KCl
 (b) $\text{C}_2\text{H}_5\text{CN}$ and 3 KCl
 (c) $\text{CH}_3\text{CH}_2\text{CONH}_2$ and 3 KCl
 (d) $\text{C}_2\text{H}_5\text{NC}$ and K_2CO_3
37. Which of the following will give N_2 gas on treatment with nitrous acid ($\text{NaNO}_2 + \text{HCl}$) ?
- (a) $\text{C}_2\text{H}_5\text{NH}_2$ (b) CH_3NH_2
 (c) $(\text{CH}_3)_2\text{CH}-\text{NH}_2$ (d) All will give N_2
38. $\text{C}_4\text{H}_{11}\text{N}$ on reaction with HNO_2 forms a tertiary alcohol. Thus, $\text{C}_4\text{H}_{11}\text{N}$ is
- (a) primary amine (b) secondary amine
 (c) tertiary amine
 (d) quaternary ammonium salt
39. The optically active compound (X)
- $$\begin{array}{c} \text{CH}_3 \\ \diagup \\ \text{C}_6\text{H}_5 \end{array} \text{CH}^*-\text{NH}_2$$
- (X)
- on treatment with NaNO_2/HCl gives
- (a) 1° alcohol with retention of configuration
 (b) 2° alcohol with inversion of configuration
 (c) racemic mixture of 2° alcohols
 (d) racemic mixture of 1° alcohols
40. Treatment of cyclobutylmethylamine with nitrous acid does not give
- (a)  (b) 
 (c)  (d) 
41. Which among the following amines can be directly oxidised to the corresponding nitro compound by potassium permanganate ?
- (a) CH_3NH_2 (b) $(\text{CH}_3)_2\text{CH}-\text{NH}_2$
 (c) $(\text{CH}_3)_2\text{NH}$ (d) $(\text{CH}_3)_3\text{C}-\text{NH}_2$
42. $\text{CH}_3\text{NH}_2 + \text{CHCl}_3 + 3\text{KOH} \longrightarrow \text{X} + \text{Y} + 3\text{H}_2\text{O}$
 compounds X and Y are :
- (a) $\text{CH}_3\text{CN} + 3\text{KCl}$ (b) $\text{CH}_3\text{NC} + 3\text{KCl}$
 (c) $\text{CH}_3\text{CONH}_2 + 3\text{KCl}$ (d) $\text{CH}_3\text{NC} + \text{K}_2\text{CO}_3$
43. By heating ammonium chloride with two equivalents of formaldehyde it forms :
- (a) dimethylamine (b) ethylamine
 (c) methylamine (d) ammonium formate
- Chemical Tests**
44. Which of the following compounds will dissolve in an alkali solution after it undergoes reaction with Hinsberg's reagent ?
- (a) CH_3NH_2 (b) $(\text{CH}_3)_3\text{N}$
 (c) $(\text{C}_2\text{H}_5)_2\text{NH}$ (d) $\text{C}_6\text{H}_5\text{NHC}_6\text{H}_5$
45. Which of the following is soluble in sodium hydroxide
- (a) 
 (b) 
 (c) 
 (d) 

46. In Hinsberg test the product of which amine does not dissolve in NaOH ?

- (a) $\text{CH}_3\text{CH}_2\text{NH}_2$ (b) $(\text{CH}_3)_2\text{CHNH}_2$
(c) CH_3NHCH_3 (d) $\text{CH}_3\text{N}(\text{CH}_3)_2$

47. Which of the following method is generally not employed for the separation of primary, secondary and tertiary amines ?

- (a) Fractional distillation
(b) Hinsberg's method
(c) Hofmann's method
(d) Filtration

Aromatic Amines

48. In the nitration of benzene using a mixture of conc. H_2SO_4 and conc. HNO_3 , the species which initiates the reaction is

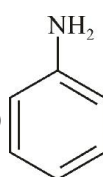
- (a) NO_2 (b) NO^+
(c) NO_2^+ (d) NO_2^-

49. Reduction of aromatic nitro compounds using Fe and HCl gives

- (a) aromatic oxime (b) aromatic hydrocarbon
(c) aromatic primary amine
(d) aromatic amide

50. The most reactive amine towards dilute hydrochloric acid is

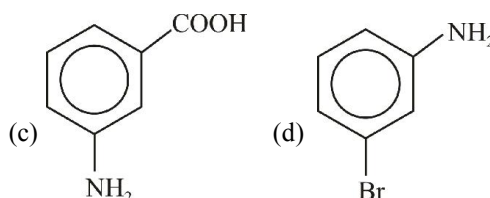
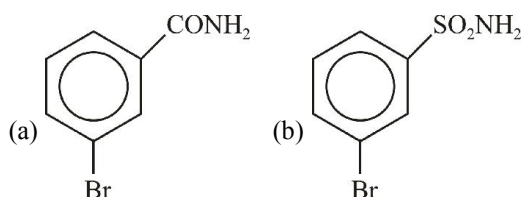
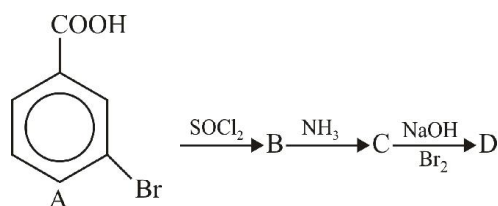
- (a) $\text{CH}_3\text{—NH}_2$ (b) $\begin{array}{c} \text{H}_3\text{C} \\ \diagup \\ \text{NH} \\ \diagdown \\ \text{H}_3\text{C} \end{array}$

- (c) $\begin{array}{c} \text{H}_3\text{C} \\ \diagup \\ \text{NH—CH}_3 \\ \diagdown \\ \text{H}_3\text{C} \end{array}$ (d) 

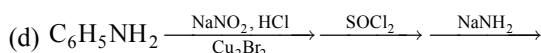
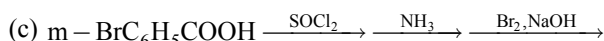
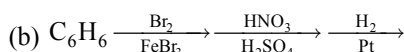
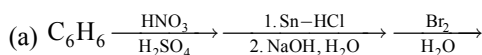
51. The reaction, $\text{ArN}_2^+\text{Cl}^- \xrightarrow{\text{Cu/HCl}} \text{ArCl} + \text{N}_2 + \text{CuCl}$ is named as

- (a) Sandmeyer reaction
(b) Gattermann reaction
(c) Claisen reaction (d) Carbylamine reaction

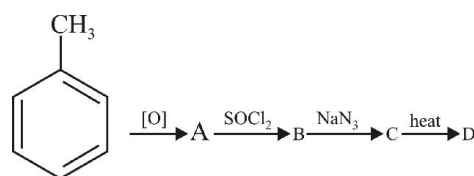
52. In a set of reactions, m-bromobenzoic acid gave a product D. Identify the product D.



53. m-Bromoaniline can be prepared by

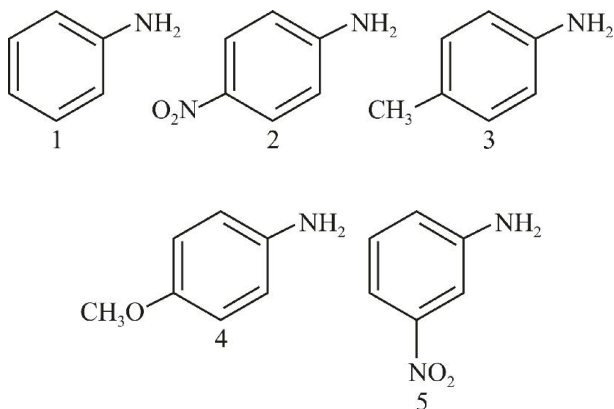


54. In the following sequence of reactions, what is D ?

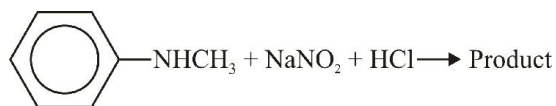


- (a) Primary amine (b) An amide
(c) Phenyl isocyanate
(d) Chain lengthened hydrocarbon

55. The correct order of increasing basic nature of the following bases is



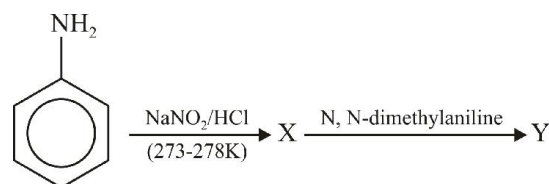
- (a) $2 < 5 < 1 < 3 < 4$ (b) $5 < 2 < 1 < 3 < 4$
 (c) $2 < 5 < 1 < 4 < 3$ (d) $5 < 2 < 1 < 4 < 3$
56. Which of the following statements about primary amines is 'False' ?
 (a) Alkylamines are stronger bases than ammonia
 (b) Alkylamines are stronger bases than aryl amines
 (c) Alkylamines react with nitrous acid to produce alcohols
 (d) Arylamines react with nitrous acid to produce phenols
57. Predict the product :



- (a) CN(C)[N+](=O)c1ccccc1
 (b) CNc1ccccc1 + ONc1ccc([N+](=O)[O-])cc1
 (c) CN(C)Oc1ccccc1
 (d) CN(C)[N+]=[N-]c1ccccc1

58. Aniline in a set of the following reactions yielded a coloured product Y.

The structure of Y would be



- (a) CN(C)c1ccc(N=Nc2ccc(NC)cc2)cc1
 (b) CN(C)c1ccc(N=Nc2ccccc2)cc1
 (c) CN(C)c1ccc(Nc2ccc(NC)cc2)cc1
 (d) Nc1ccc(N=Nc2ccc(C)cc2)cc1

59. Aniline on heating with fuming sulphuric acid gives :

- (a) aniline sulphate (b) sulphanilic acid
 (c) aniline 2,4-disulphonic acid
 (d) nitrobenzene

60. Aniline on treatment with aqueous bromine gives :

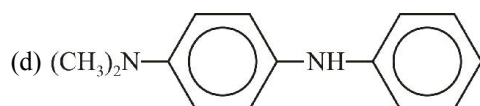
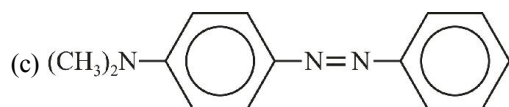
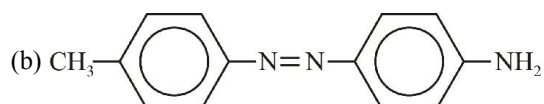
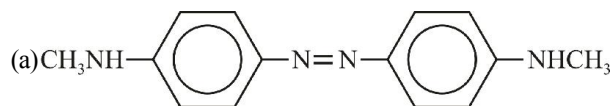
- (a) 2,4,6-tribromo aniline
 (b) o-bromo aniline
 (c) 2,4-dibromo aniline
 (d) p-bromo aniline

61. X is
- (a) Na2S (b) Sn/HCl
 (c) LiAlH4 (d) all of these

62. Reaction of aniline with benzaldehyde is :

- (a) substitution (b) addition
(c) condensation (d) polymerisation

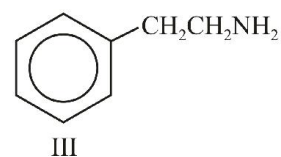
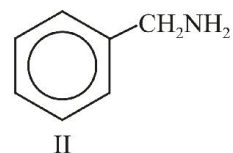
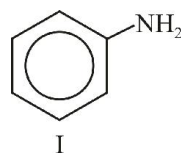
63. Aniline when diazotised in cold and then treated with dimethyl aniline gives a coloured product. Its structure would be :



64. Which of the following reagents can be used to prepare ethyl carbylamine from ethyl iodide ?

- (a) KCN (b) AgCN
(c) CuCN (d) HCN

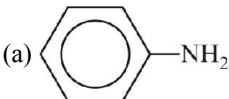
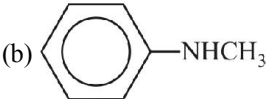
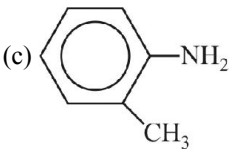
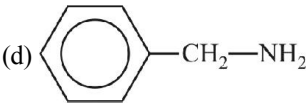
65. Consider the following compounds,



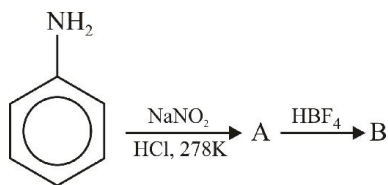
What is the correct order of basicity of the above compounds ?

- (a) I > II > III (b) III > I > II
(c) III > II > I (d) I > III > II

EXERCISE - 2 : PREVIOUS YEAR JEE MAINS QUESTION

1. On heating benzyl amine with chloroform and ethanolic KOH, product obtained is (2002)
(a) benzyl alcohol (b) benzaldehyde
(c) benzonitrile (d) benzeneisocyanide
2. The correct order of increasing basic nature for the bases NH_3 , CH_3NH_2 and $(\text{CH}_3)_2\text{NH}$ is (2003)
(a) $\text{CH}_3\text{NH}_2 < \text{NH}_3 < (\text{CH}_3)_2\text{NH}$
(b) $(\text{CH}_3)_2\text{NH} < \text{NH}_3 < \text{CH}_3\text{NH}_2$
(c) $\text{NH}_3 < \text{CH}_3\text{NH}_2 < (\text{CH}_3)_2\text{NH}$
(d) $\text{CH}_3\text{NH}_2 < (\text{CH}_3)_2\text{NH} < \text{NH}_3$
3. The ammonia evolved from the treatment of 0.30 g of an organic compound for the estimation of nitrogen was passed in 100 mL of 0.1 M sulphuric acid. The excess of acid required 20 mL of 0.5 M sodium hydroxide solution for complete neutralisation. The organic compound is (2004)
(a) acetamide (b) benzamide
(c) urea (d) thiourea
4. Which of the following is the strongest base ? (2004)
(a)  (b) 
(c)  (d) 
5. An organic compound having molecular mass 60 is found to contain C = 20%, H = 6.67% and N = 46.67% while rest is oxygen. On heating it gives NH_3 alongwith a solid residue. The solid residue gives violet colour with alkaline copper sulphate solution. The compound is (2005)
(a) $\text{CH}_3\text{CH}_2\text{CONH}_2$ (b) $(\text{NH}_2)_2\text{CO}$
(c) CH_3CONH_2 (d) CH_3NCO
6. Reaction of cyclohexanone with dimethylamine in the presence of catalytic amount of an acid forms a compound. Water during the reaction is continuously removed. The compound formed is generally known as (2005)
(a) an amine (b) an imine
(c) an enamine (d) a Schiff's base
7. Amongst the following the most basic compound is (2005)
(a) p-nitroaniline (b) acetanilide
(c) aniline (d) benzylamine
8. Which one of the following methods is neither meant for the synthesis nor for separation of amines ? (2005)
(a) Curtius reaction (b) Wurtz reaction
(c) Hofmann method (d) Hinsberg method
9. Fluorobenzene ($\text{C}_6\text{H}_5\text{F}$) can be synthesized in the laboratory (2006)
(a) by heating phenol with HF and KF
(b) from aniline by diazotisation followed by heating the diazonium salt with HBF_4
(c) by direct fluorination of benzene with F_2 gas
(d) by reacting bromobenzene with NaF solution
10. In the chemical reaction, (2007)
 $\text{CH}_3\text{CH}_2\text{NH}_2 + \text{CHCl}_3 + 3\text{KOH} \longrightarrow (\text{A}) + (\text{B}) + 3\text{H}_2\text{O}$
the compounds (A) and (B) are respectively
(a) $\text{C}_2\text{H}_5\text{CN}$ and 3KCl
(b) $\text{CH}_3\text{CH}_2\text{CONH}_2$ and 3KCl
(c) $\text{C}_2\text{H}_5\text{NC}$ and K_2CO_3 (d) $\text{C}_2\text{H}_5\text{NC}$ and 3KCl
11. Which one of the following is the strongest base in aqueous solution ? (2007)
(a) Trimethylamine (b) Aniline
(c) Dimethylamine (d) Methylamine
12. Toluene is nitrated and the resulting product is reduced with tin and hydrochloric acid. The product so obtained is diazotised and then heated with cuprous bromide. The reaction mixture so formed contains (2008)
(a) mixture of o- and p-bromotoluenes
(b) mixture of o- and p-dibromobenzenes
(c) mixture of o- and p-bromoanilines
(d) mixture of o- and m-bromotoluenes

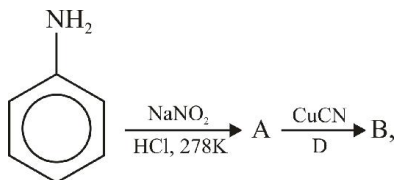
13. In the chemical reactions,



the compounds 'A' and 'B' respectively are (2010)

- (a) nitrobenzene and fluorobenzene
- (b) phenol and benzene
- (c) benzene diazonium chloride and fluorobenzene
- (d) nitrobenzene and chlorobenzene

14. In the chemical reactions



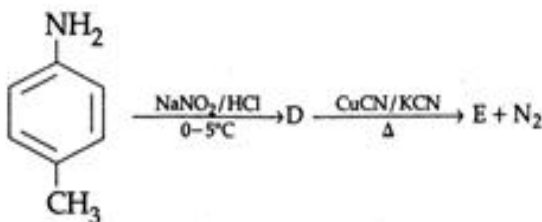
Compounds A and B respectively are (2011)

- (a) fluorobenzene and phenol
- (b) benzene diazonium chloride and benzonitrile

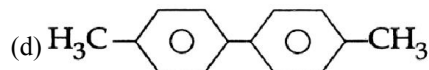
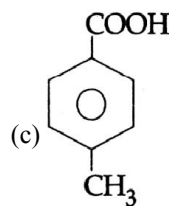
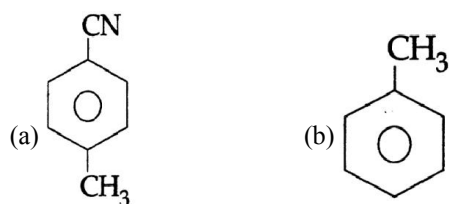
15. On heating an aliphatic primary amine with chloroform and ethanolic potassium hydroxide, the organic compound formed is : (2014)

- (a) an alkanediol
- (b) an alkyl cyanide
- (c) an alkyl isocyanide
- (d) an alkanol

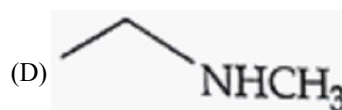
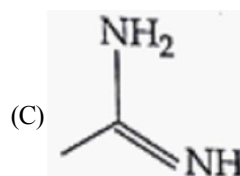
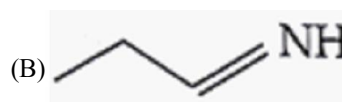
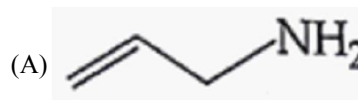
16. In the reaction



the product E is: (2015)



17. The increasing order of basicity of the following compounds is : (2018)



- (a) (D) < (B) < (A) < (C)
- (b) (A) < (B) < (C) < (D)
- (c) (B) < (A) < (C) < (D)
- (d) (B) < (A) < (D) < (C)

JEE MAINS ONLINE QUESTION

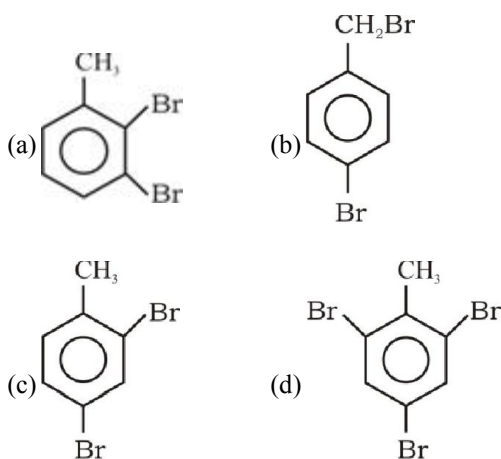
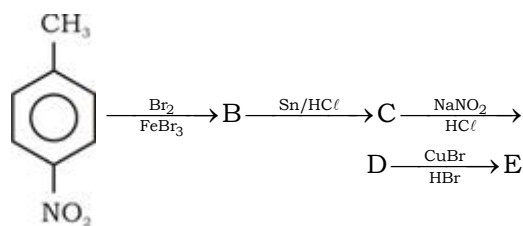
1. Complete reduction of benzene-diazonium chloride with Zn/HCl gives: **Online 2014 SET (2)**

- (a) Phenylhydrazine
- (b) Aniline
- (c) Hydrazobenzene
- (d) Azobenzene

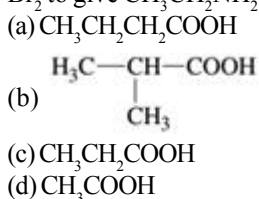
2. The final product formed when Methyl amine is treated with NaNO_2 and HCl is : **Online 2014 SET (4)**

- (a) Nitromethane
- (b) Diazomethane
- (c) Methylcyanide
- (d) Methylalcohol

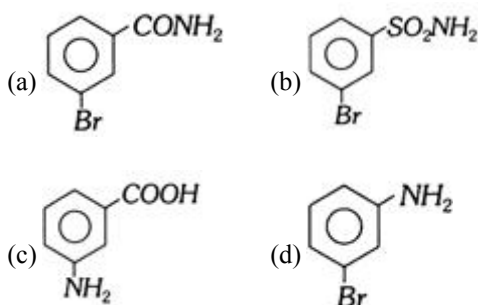
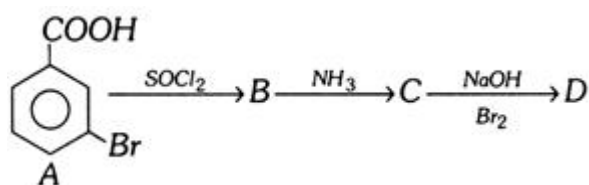
3. In a set of reaction p-nitrotoluene yielded a product E
Online 2014 SET (4)



4. An organic compound A upon reacting with NH_3 gives B. On heating B gives C. C in presence of KOH reacts with Br_2 to give $\text{CH}_3\text{CH}_2\text{NH}_2$. A is
Online 2015 SET (1)



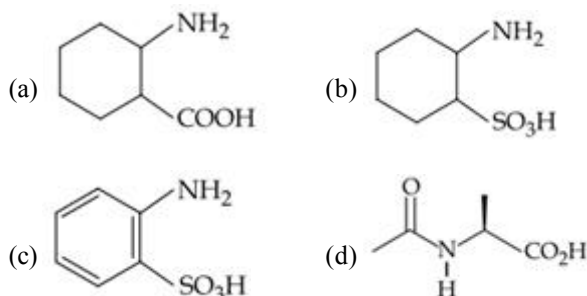
5. In a set of reactions m-bromobenzoic acid gave a product D. Identify the product D
Online 2015 SET (1)



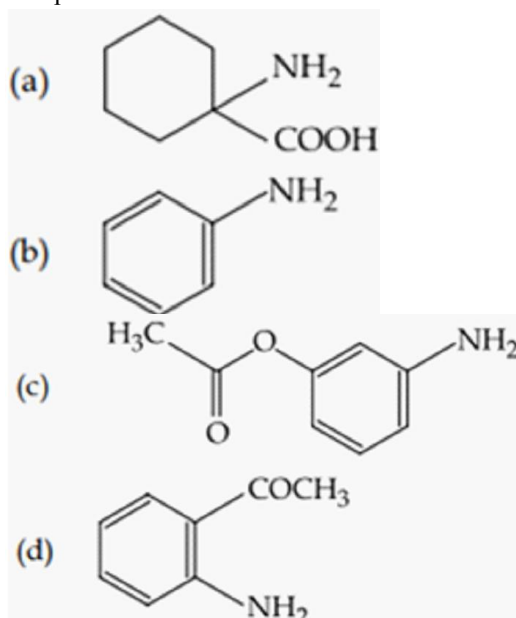
6. The test to distinguish primary, secondary and tertiary amines is :
Online 2016 SET (1)

(a) Carbylamine reaction
(b) $\text{C}_6\text{H}_5\text{SO}_2\text{Cl}$
(c) Sandmeyer's reaction
(d) Mustard oil test

7. Which of the following will not exist in zwitter ionic form at $\text{pH} = 7$?
Online 2018 SET (1)



8. The increasing order of diazotisation of the following compounds is :
Online 2018 SET (2)



9. Which of the following amines can be prepared by Gabriel phthalimide reaction ?
Online 2019 SET (1)

(a) t-butylamine (b) n-butylamine
(c) neo-pentylamine (d) triethylamine

10. Hinsberg's reagent is :
Online 2019 SET (3)

(a) $\text{C}_6\text{H}_5\text{SO}_2\text{Cl}$ (b) $\text{C}_6\text{H}_5\text{COCl}$
(c) SOCl_2 (d) $(\text{COCl})_2$

EXERCISE - 3 : ADVANCED OBJECTIVE QUESTIONS

Properties and Nomenclature

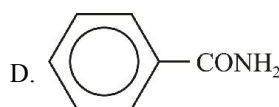
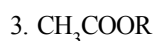
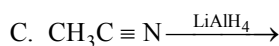
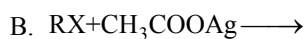
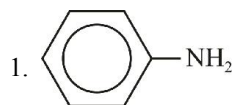
1. (S) In the explosive amatol, TNT is mixed with
(a) Ammonium citrate
(b) Ammonium nitrate
(c) Ammonium oxalate
(d) Ammonium sulphate
2. (S) Correct order of increasing basicity is
(a) $\text{NH}_3 < \text{C}_6\text{H}_5\text{NH}_2 < (\text{C}_2\text{H}_5)_2\text{NH} < \text{C}_2\text{H}_5\text{NH}_2 < (\text{C}_2\text{H}_5)_3\text{N}$
(b) $\text{C}_6\text{H}_5\text{NH}_2 < \text{NH}_3 < (\text{C}_2\text{H}_5)_3\text{N} < (\text{C}_2\text{H}_5)_2\text{NH} < \text{C}_2\text{H}_5\text{NH}_2$
(c) $\text{C}_6\text{H}_5\text{NH}_2 < \text{NH}_3 < \text{C}_2\text{H}_5\text{NH}_2 < (\text{C}_2\text{H}_5)_3\text{N} < (\text{C}_2\text{H}_5)_2\text{NH}$
(d) $\text{C}_6\text{H}_5\text{NH}_2 < (\text{C}_2\text{H}_5)_3\text{N} < \text{NH}_3 < \text{C}_2\text{H}_5\text{NH}_2 < (\text{C}_2\text{H}_5)_2\text{NH}$
3. (S) Which statement is not correct
(a) Amines form hydrogen bond
(b) Ethyl amine has higher boiling point than propane
(c) Methyl amine is more basic than ammonia
(d) Dimethyl amine is less basic than methyl amine
4. (S) How many isomeric amines having the formula $\text{C}_3\text{H}_9\text{N}$ are possible ?
(a) Two (b) Three
(c) Four (d) Five
5. (S) Aniline is a weaker base than ethylamine because
(a) The phenyl group in aniline increases the electron density on the nitrogen atom.
(b) The ethyl group in ethylamine decreases the electron density on the nitrogen atom.
(c) The lone pair of electrons on the nitrogen atom in aniline is delocalized over the benzene ring.
(d) Aniline is less soluble in water than ethylamine.
6. (S) A solution of ethylamine :
(a) turns blue litmus red
(b) turns red litmus blue
(c) does not affect the litmus
(d) bleaches the litmus
7. (S) The hybridization of nitrogen atom in amines is :
(a) sp (b) sp^2
(c) sp^3 (d) dsp^2
8. (S) Trimethylamine has :
(a) planar geometry
(b) trigonal bipyramidal geometry
(c) pyramidal shape (d) octahedral geometry
9. (S) The boiling points of amines and their corresponding alcohols and acids vary in the order :
(a) $\text{RCH}_2\text{NH}_2 > \text{RCOOH} > \text{RCH}_2\text{OH}$
(b) $\text{RCH}_2\text{NH}_2 > \text{RCH}_2\text{OH} > \text{RCOOH}$
(c) $\text{RCH}_2\text{NH}_2 < \text{RCOOH} < \text{RCH}_2\text{OH}$
(d) $\text{RCH}_2\text{NH}_2 < \text{RCH}_2\text{OH} < \text{RCOOH}$
10. (S) Which has the highest pK_b value ?
(a) $\text{R}_3\text{C}-\text{NH}_2$ (b) R_2NH
(c) RNH_2 (d) NH_3
11. (M) The name $\text{CH}_3-\text{C}\equiv\text{N}$ is :
(a) methyl cyanide (b) methyl carbylamine
(c) ethanenitrile (d) acetonitrile
12. (A) **Assertion :** n-Propylamine has a higher boiling point than trimethylamine.
Reason : Among n-propylamine molecules there is hydrogen bonding, but there is no hydrogen bonding among trimethylamine molecules.
(a) A (b) B
(c) C (d) D
- Preparation of Aliphatic Amines**
13. (S) $\text{CH}_3\text{CN} \xrightarrow{\text{Na} + \text{C}_2\text{H}_5\text{OH}} \text{X}$
The compound X is
(a) CH_3CONH_2 (b) $\text{CH}_3\text{CH}_2\text{NH}_2$
(c) C_2H_6 (d) CH_3NHCH_3

14. (S) Which of the following reactions does not yield an amine
- (a) $RX + NH_3 \longrightarrow$
- (b) $RCH=NOH + [H] \xrightarrow[C_2H_5OH]{Na}$
- (c) $RCN + H_2O \xrightarrow{H^+}$
- (d) $RCNH_2 + 4H \xrightarrow{LiAlH_4}$
15. (S) (A) $\xrightarrow{H_2/Pt} 1^\circ$ Amine
- (B) $\xrightarrow{H_2/Pt} 2^\circ$ Amine
- (A) and (B) respectively are :
- (a) RNC, RNC (b) RCN, RCN
- (c) RCN, RNC (d) RNC, RCN
16. (M) (A) $\xrightarrow[H]{Reduction} \text{Primary amine}$
- The compound (A) may be :
- (a) alkyl isocyanide (b) alkyl cyanide
- (c) acidamide (d) 1° nitroalkane
17. (M) A mixture of three amines (pri., sec. and tert.) can be prepared by :
- (a) Hofmann's bromamide reaction
- (b) the reduction of nitroalkanes, cyanides and oximes
- (c) ammonolysis of alkylhalides.
- (d) ammonolysis of alcohols
18. (M) Primary amines may be obtained by :
- (a) the reduction of nitroalkanes
- (b) the reduction of alkyl isocyanide
- (c) the decarboxylation of amino acids
- (d) the hydrolysis of alkyl isocyanate
19. (M) Tertiary amines may be obtained by :
- (a) Gabriel phthalimide synthesis
- (b) heating an alcoholic solution of ammonia with excess of RX
- (c) the hydrolysis of dialkyl cyanamide
- (d) thermal decomposition of quaternary ammonium hydroxide
20. (M) When methyl iodide is treated with ammonia, the product obtained is/are :
- (a) methylamine
- (b) dimethylamine
- (c) trimethylamine
- (d) quaternary ammonium salt
21. (A) **Assertion :** In order to convert R-Cl to pure R-NH₂, Gabriel phthalimide synthesis can be used.
- Reason :** With proper choice of alkyl halides, phthalimide synthesis can be used to prepare 1° , 2° or 3° amines.
- (a) A (b) B
- (c) C (d) D
- Reactions of Aliphatic Amines**
22. (S) C₃H₉N represents
- (a) Primary amine (b) Secondary amine
- (c) Tertiary amine (d) All
23. (S) Ethyl amine on heating with CS₂ in presence of HgCl₂ forms
- (a) C₂H₅NCS (b) (C₂H₅)₂S
- (c) (C₂H₅)₂CS (d) C₂H₅(CS)₂
24. (S) Ethyl amine undergoes oxidation in the presence of KMnO₄ to form
- (a) An acid (b) An alcohol
- (c) An aldehyde (d) A nitrogen oxide
25. (S) Which of the following on reduction with hydrogen and Raney nickel gives benzylamine ?
- (a) Benzonitrile (b) Acetonitrile
- (c) Propionitrile (d) Butyronitrile
26. (S) Which of the following amides will not undergo Hofmann bromamide reaction ?
- (a) CH₃CONH₂ (b) CH₃CH₂CONH₂
- (c) C₆H₅CONH₂ (d) CH₃CONHCH₃
27. (S) Ethylamine reacts with nitrous acid to form :
- (a) methyl alcohol (b) ethyl alcohol
- (c) ethane (d) ethyl nitrite

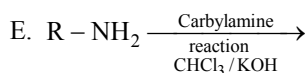
28. (S) Match the List I with List II and pick the correct matching from the codes given below :

List - I
(Reactants)

List - II
(Products)



$\xrightarrow{\text{Hofmann's degradation}}$



Codes :

(a) A-3, B-1, C-2, D-5, E-4

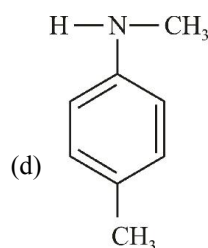
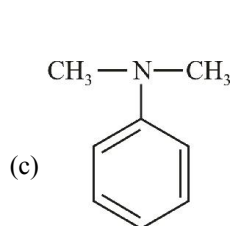
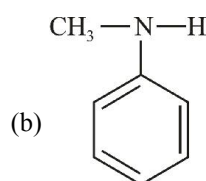
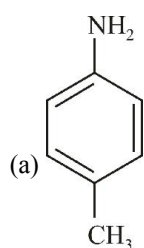
(b) A-2, B-3, C-4, D-1, E-5

(c) A-3, B-2, C-1, D-5, E-4

(d) A-5, B-3, C-2, D-4, E-1

(e) A-4, B-2, C-3, D-5, E-1

29. (S) Which of the following amines will show carbylamine reaction ?



30. (M) All the three amines (pri., sec. and tert.) react with :

(a) HOH

(b) RX

(c) HCl

(d) CH_3COCl

31. (M) $C_4H_{11}N(X) + HNO_2 \rightarrow C_4H_{10}O$ (3° alcohol) hence, (X) will give :

(a) Carbylamine reaction

(b) Hofmann's mustard oil reaction

(c) Diazonium salt (as the intermediate)

(d) Hofmann's bromamide reaction

Comprehension

The most important reaction of all kind of amines is their reaction with nitrous acid. It is also a distinguish test as primary, secondary, tertiary, aromatic amines give different products on reaction with nitrous acid. Primary amines react with nitrous acid to give alcohols as the major product along with some other minor product like alkene etc. In case of aromatic amines the reaction is called diazotization and the product obtained is of great synthetic importance. In case of some cyclic primary amines ring expansion and contraction is also observed.

32. (C) When propanamine reacts with nitrous acid which of the following statements are true ?

(I) Here reaction intermediate is secondary carbocation.

(II) Here the product formed are propene, 2 chloropropane and propan-2-ol.

(III) It involves hydride shift.

(IV) Here one of the product is optically active also.

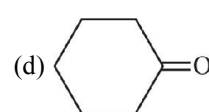
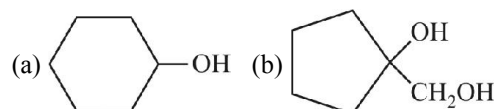
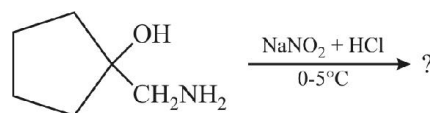
(a) I, II, III

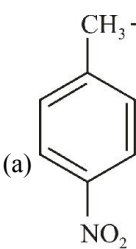
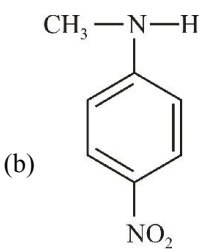
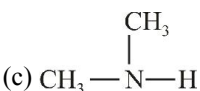
(b) I, II

(c) I, II, IV

(d) II, III, IV

33. (C) The major product obtained in this reaction can be given as



- 34. (C)** Which of these compound can give an N-nitroso amine on reaction with nitrous acid ?
 (I) $C_6H_5CONHC_2H_5$ (II) $C_2H_5NHC_2H_5$
 (III) $C_6H_5NHC_2H_5$ (IV) $C_6H_5NHC_6H_{11}$
 (a) I, II, III (b) I, III, IV
 (c) II, III, IV (d) I, II, III, IV
- 35. (A)** **Assertion :** In Hofmann bromide reaction, the amine formed has one carbon atom less than the parent 1° amide.
Reason : N-methyl acetamide undergoes Hofmann bromamide reaction.
 (a) A (b) B
 (c) C (d) D
- 36. (A)** **Assertion :** Carbylamine reaction involves the reaction between 1° amine and chloroform in basic medium.
Reason : In carbylamine reaction, $-NH_2$ group is converted into $-NC$ group.
 (a) A (b) B
 (c) C (d) D
- 37. (X)** **Column I** **Column II**
 (A) Gabriel phthalimide reaction (p) $C_6H_5CH_2NH_2$
 (B) Reduction product of nitrogen containing (q) $C_6H_5NH_2$
 (C) Reaction product of amines with alcoholic KOH and $CHCl_3$ (r) C_6H_5NC
 (D) Reaction product of 1° amides with $Br_2 + KOH$ (s) $CH_3CH_2NH_2$
- Chemical Tests**
- 38. (M)** Reaction of $RCONH_2$ with a mixture of Br_2 and KOH gives RNH_2 as the main product. The intermediates involved in the reaction are :
 (a) $RCONHBr$ (b) $R-NHBr$
 (c) $R-N=C=O$ (d) $RCONBr_2$
- 39. (S)** Azo dye test is given by
 (a) All amines
 (b) Only secondary amines
 (c) Only primary aliphatic amines
 (d) Only primary aromatic amines
- 40. (S)** A mixture of 1° , 2° and 3° amines can be separated by Hinsberg's reagent which is
 (a) Benzoyl chloride
 (b) Acetyl chloride
 (c) Benzenesulphonyl chloride
 (d) Benzyl chloride
- 41. (S)** Which of the following reacts with Hinsberg's reagent (Benzene sulphonyl chloride) to form the product soluble in KOH ?
 (a) Primary amine (b) Secondary amine
 (c) Tertiary amine (d) Quaternary amine
- 42. (S)** Hinsberg's reagent is :
 (a) benzene sulphonamide
 (b) benzene sulphonic acid
 (c) benzene sulphuryl chloride
 (d) benzene sulphonyl chloride
- 43. (S)** Primary amines are identified by :
 (a) Hofmann's reaction
 (b) Carbylamine reaction
 (c) Friedel-Crafts reaction
 (d) Biuret reaction
- 44. (M)** Which of the following amines will show positive Liebermann's nitroso test :
 (a)  (b) 
 (c)  (d) None of these
- 45. (M)** Primary, secondary and tertiary amines can be distinguished by the action of :
 (a) $CS_2/HgCl_2$ (b) $NaNO_2/HCl$
 (c) $CHCl_3/KOH$ (d) $X_2/NaOH$

46. (M) A mixture of three amines can be separated into individual one by :

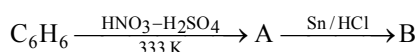
- (a) Carbylamine reaction
- (b) Hinsberg's method
- (c) Fractional distillation method
- (d) Hofmann's method

Aromatic Amines

47. (S) In the diazotisation of aniline with sodium nitrite and hydrochloric acid, an excess of hydrochloric acid is used primarily to

- (a) Suppress the concentration of free aniline available for coupling
- (b) Suppress hydrolysis of phenol
- (c) Insure a stoichiometric amount of nitrous acid
- (d) Neutralize the base liberated

48. (S) What is the end product in the sequence of following reactions ?

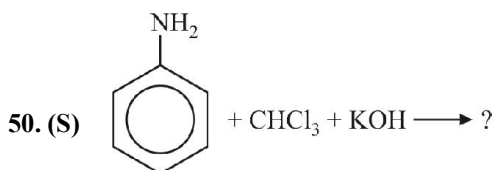


- (a) Nitrobenzene (b) Aniline
- (c) Benzonitrile (d) Phenyl carbylamine

49. (S) In the reaction $\text{C}_6\text{H}_5\text{CHO} + \text{C}_6\text{H}_5\text{NH}_2 \rightarrow \text{C}_6\text{H}_5\text{N}=\text{CHC}_6\text{H}_5 + \text{H}_2\text{O}$,

the compound $\text{C}_6\text{H}_5\text{N}=\text{CHC}_6\text{H}_5$ is known as

- (a) Aldol (b) Schiff's reagent
- (c) Schiff's base (d) Benedict reagent



- (a) Phenyl isocyanide (b) Benzyl amine
- (c) Benzyl chloride (d) None of these

51. (A) **Assertion :** Benzene diazonium chloride does not give tests for nitrogen.

Reason : N_2 gas is lost during heating.

- (a) A (b) B
- (c) C (d) D

52. (A) **Assertion :** p-Anisidine is weaker base than aniline.

Reason : $-\text{OCH}_3$ group in anisidine exerts $-\text{R}$ effect.

- (a) A (b) B
- (c) C (d) D

53. (A) **Assertion :** In strongly acidic solutions, aniline becomes more reactive towards electrophilic reagents.

Reason : The amino group being completely protonated in strongly acidic solution, the lone pair of electrons on the nitrogen is no longer available for resonance.

- (a) A (b) B
- (c) C (d) D

54. (A) **Assertion :** Aniline hydrogen sulphate, on heating, forms a mixture of o-and p-amino benzene sulphonic acid.

Reason : The sulphonic acid group is electron withdrawing.

- (a) A (b) B
- (c) C (d) D

55. (A) **Assertion :** Amines are more basic than esters and ethers.

Reason : Nitrogen is less electronegative than oxygen. It is in better position to accommodate the positive charge on the proton

- (a) A (b) B
- (c) C (d) D

56. (A) **Assertion :** In Sandmeyer reaction, diazonium salt is heated with copper powder.

Reason : Sandmeyer reaction occurs by a free radical mechanism.

- (a) A (b) B
- (c) C (d) D

57. (A) **Assertion :** Me_3N reacts with BF_3 whereas Ph_3N does not.

Reason : The electron pair on nitrogen atom in Ph_3N is delocalised in the benzene ring and is not available to boron in BF_3 .

- (a) A (b) B
- (c) C (d) D

58. (A) **Assertion :** Aniline hydrogen sulphate on heating forms a mixture of ortho and para aminobenzene sulphonic acids.

Reason : The sulphonic acid group is electron withdrawing.

- (a) A (b) B
(c) C (d) D

59. (A) **Assertion :** Aniline is a weaker base than benzylamine.

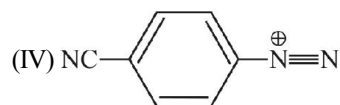
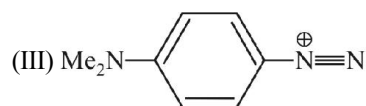
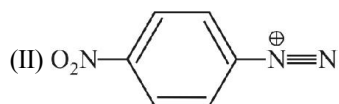
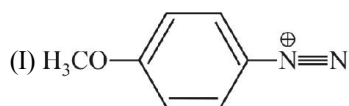
Reason : In aniline, mesomeric interaction occurs between benzene ring and amino group.

- (a) A (b) B
(c) C (d) D

Comprehension

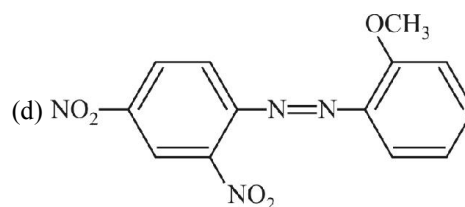
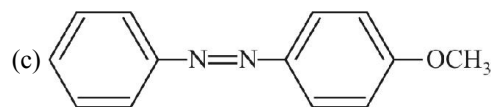
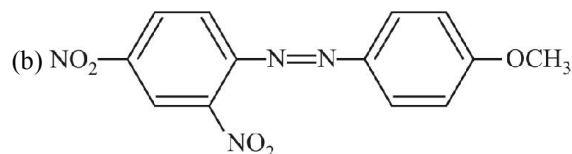
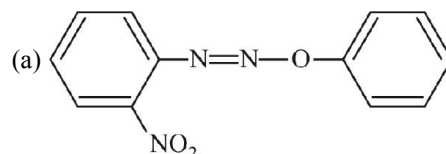
When an aromatic primary amine is treated with $\text{NaNO}_2 + \text{HCl}$ at $0-5^\circ\text{C}$, a diazonium salt is formed and the reaction is called diazo reaction. In this reaction mineral acid must be added to prevent the coupling reaction of diazonium salt with excess of aryl amine. The diazonium salt is very widely used in the synthesis of a number of organic compounds. Its coupling reaction is highly useful in the synthesis of a number of coloured dyes.

60. (C) For the following diazonium ions the decreasing order of reactivity of these ions in azo-coupling reactions (under similar conditions) can be given as

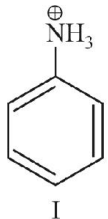


- (a) II > IV > III > I (b) II > IV > I > III
(c) II > I > IV > III (d) II > III > IV > I

61. (C) When 2, 4-dinitro aniline reacts with $\text{NaNO}_2 + \text{HCl}$ at 5°C followed by reaction with anisole, a coloured compound is formed which can be given as

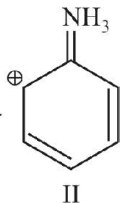


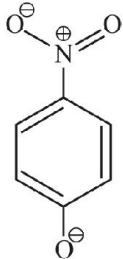
EXERCISE - 4 : PREVIOUS YEAR JEE ADVANCED QUESTION

1. The compound which on reaction with aqueous nitrous acid at low temperature produces an oily nitrosoamine, is
(a) methylamine (b) ethylamine (1981)
(c) diethylamine (d) triethylamine
2. Acetamide is treated separately with the following reagents. Which one of these would give methylamine ?
(a) PCl_5 (b) Sodalime (1983)
(c) $\text{NaOH} + \text{Br}_2$ (d) Hot conc. H_2SO_4
3. Carbylamine test is performed in alc. KOH by heating a mixture of (1984)
(a) chloroform and silver powder
(b) trihalogenated methane and a primary amine
(c) an alkyl halide and a primary amine
(d) an alkyl cyanide and a primary amine
4. Examine the following two structures for the anilinium ion and choose the correct statement from the ones given below (1993)
- 

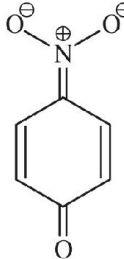
I

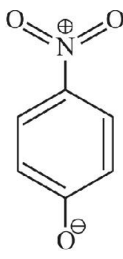
$\longleftrightarrow$



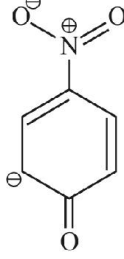
II
- (a) II is not an acceptable canonical structure because carbonium ions are less stable than ammonium ions
(b) II is not an acceptable canonical structure because it is non-aromatic
(c) II is not an acceptable canonical structure because the nitrogen has 10 valence electrons
(d) II is an acceptable canonical structure
5. The most unlikely representation of resonance structures of p-nitrophenoxide ion is (1999)
- 

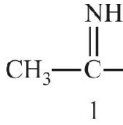
(a)



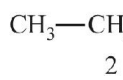
(b)
- 

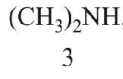
(c)



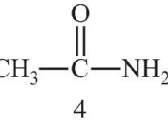
(d)
6. Among the following, the strongest base is (2000)
(a) $\text{C}_6\text{H}_5\text{NH}_2$ (b) $\text{p-NO}_2\text{C}_6\text{H}_4\text{NH}_2$
(c) $\text{m-NO}_2\text{C}_6\text{H}_4\text{NH}_2$ (d) $\text{C}_6\text{H}_5\text{CH}_2\text{NH}_2$
7. The correct order of the basicities of the following compounds is (2001)
- 

1

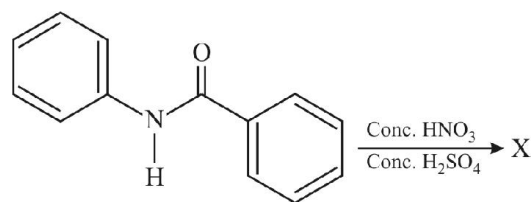
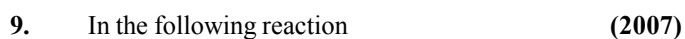
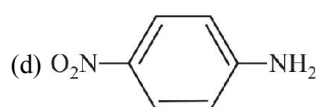
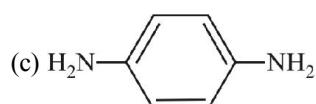
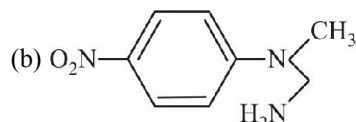
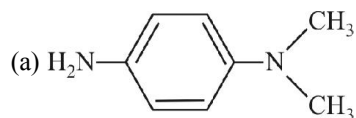
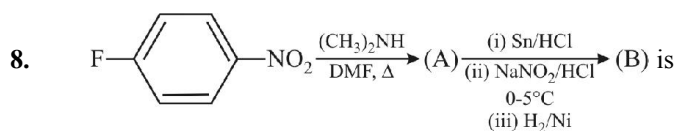


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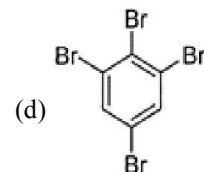
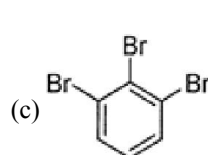
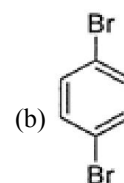
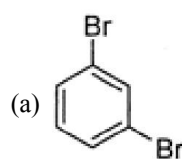
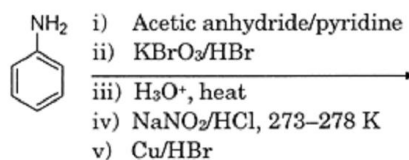
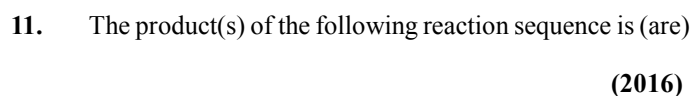
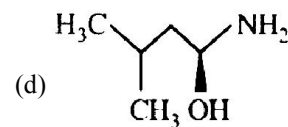
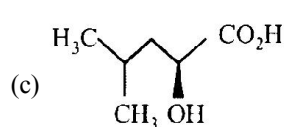
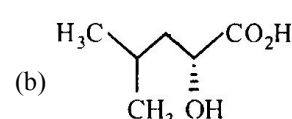
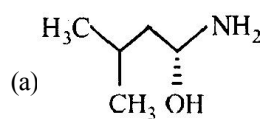
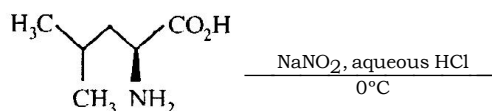
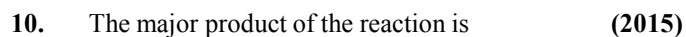
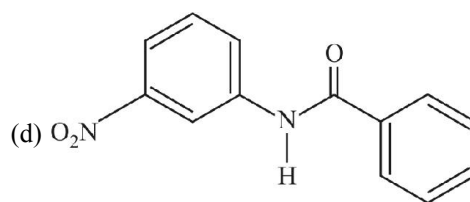
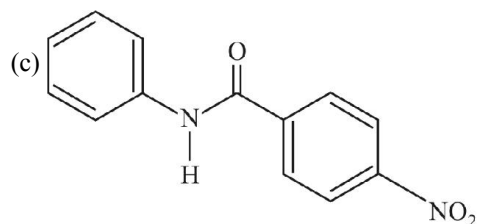
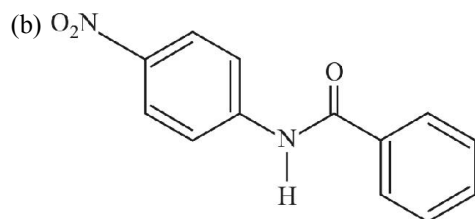
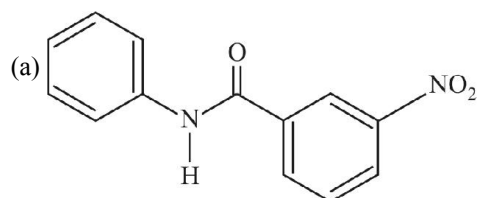
3



4
- (a) $2 > 1 > 3 > 4$ (b) $1 > 3 > 2 > 4$
(c) $3 > 1 > 2 > 4$ (d) $1 > 2 > 3 > 4$



The structure of the major product X is



Multiple Answer Questions

12. When nitrobenzene is treated with Br_2 in presence of FeBr_3 , the major product formed is m-bromonitrobenzene. Statements which are related to obtain the m-isomer, are

(1992)

- (a) The electron density on meta-carbon is more than that on ortho and para-positions.
- (b) The intermediate carbonium ion formed after initial attack of $\text{Br}^\oplus$ at the meta-position is least destabilised.
- (c) Loss of aromaticity when $\text{Br}^\oplus$ attacks at the ortho and para-positions and not at meta-position.
- (d) Easier loss of $\text{H}^\oplus$ to regain aromaticity from the meta-position than from ortho and para-positions.

13. p-chloroaniline and anilinium hydrochloride can be distinguished by

(1998)

- (a) Sandmeyer reaction
- (b) NaHCO_3
- (c) AgNO_3
- (d) Carbylamine test

14. A positive carbylamine test is given by

(1999)

- (a) N, N-dimethyl aniline
- (b) 2, 4-dimethyl aniline
- (c) N-methyl-o-methyl aniline
- (d) p-methyl benzylamine

15. Hydrogen bonding plays a central role in the following phenomena :

(2014)

- (a) Ice floats in water
- (b) Higher Lewis basicity of primary amines than tertiary amines in aqueous solutions.
- (c) Formic acid is more acidic than acetic acid.
- (d) Dimerisation of acetic acid in benzene.

Assertion Reason

- (A) If both ASSERTION and REASON are true and reason is the correct explanation of the assertion.
- (B) If both ASSERTION and REASON are true but reason is not the correct explanation of the assertion.

- (C) If ASSERTION is true but REASON is false.
- (D) If both ASSERTION and REASON are false.
- (E) If ASSERTION is false but REASON is true.

16. **Assertion :** In strongly acidic solutions, aniline becomes more reactive towards electrophilic reagents.

Reason : The amino group being completely protonated in strongly acidic solution, the lone pair of electrons on the nitrogen is no longer available for resonance. (2001)

17. **Assertion :** Aniline on reaction with NaNO_2/HCl at 0°C followed by coupling with β -naphthol gives a dark blue coloured precipitate.

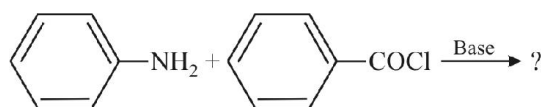
Reason : The colour of the compound formed in the reaction of aniline with NaNO_2/HCl at 0°C followed by coupling with β -naphthol is due to the extended conjugation. (2008)

Fill in the Blanks

18. Amongst the three isomers of nitrophenol, the one that is least soluble in water is (1992)
19. The high melting point and insolubility in organic solvents of sulphanilic acid are due to its structure. (1994)

Subjective Questions

20. State the equations for the preparation of following compounds. (Equations need not to be balanced) (1982)
- (i) Chlorobenzene from aniline (in two steps).
 - (ii) n-propyl amine from ethyl chloride (in two steps).
21. State the conditions under which the following preparation is carried out. Give the necessary equations which need not to be balanced. (1983)
- “Aniline from benzene”.
22. How would you convert ? (1985)
- “Aniline to chlorobenzene.”
23. For nitromethane molecule, write structure(s) (1986)
- (i) showing significant resonance stabilisation
 - (ii) indicating tautomerism
24. Complete the following with appropriate structures (1986)



25. Give a chemical test and the reagents used to distinguish between the following : (1988)

“Ethylamine and diethylamine”.

26. Arrange the following in increasing order of base strength methylamine, dimethylamine, aniline, N-methylaniline.

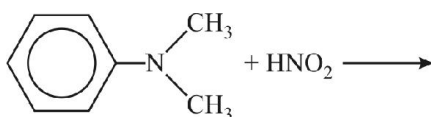
(I) (II) (III) (IV)

(1988)

27. How will you bring about the following conversion ?

“4-nitro aniline to 1,2,3-tribromo benzene.” (1990)

28. Write the structure of the major organic product expected from the following reaction. (1992)



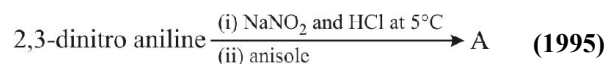
29. A basic volatile nitrogen compound gave a foul smelling gas when treated with chloroform and alcoholic potash. A 0.295 g sample of the substance dissolved in aq. HCl and treated with NaNO₂. Solution at 0°C, liberated a colourless, odourless gas whose volume corresponded to 112 mL at STP. After the evolution of the gas was complete, the aqueous solution was distilled to give an organic liquid which did not contain nitrogen and which on warming with alkali and iodine gave a yellow precipitate. Identify the original substance. Assuming that it contains one N-atom per molecule. (1993)

30. Outline a synthesis of p-bromonitrobenzene from benzene in two steps. (1993)

21. Give the structure of A, (explanations are not required.)

“A (C₃H₉N) reacts with benzenesulphonyl chloride to give a solid, insoluble in alkali.” (1993)

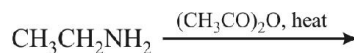
32. Complete the following with appropriate structure :



33. Write the structure of the foul smelling compound obtained when aniline is treated with chloroform in the presence of KOH. (1996)

34. Give reasons for the following in one or two sentences : “Dimethyl amine is a stronger base than trimethyl amine.” (1998)

35. Following reaction gives two products. Write the structures of the products. (1998)

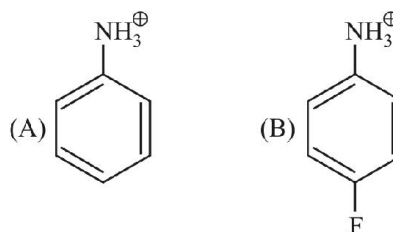


36. How would you bring about the following conversion (in 3 steps) (2000)

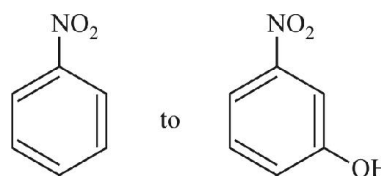
Aniline $\longrightarrow$ Benzylamine

37. There is a solution of p-hydroxybenzoic acid and p-amino benzoic acid. Discuss one method by which we can separate them and also write down the confirmatory test of the functional groups present. (2003)

38. Which of the following is more acidic and why ? (2004)



39. Convert (2004)



in not more than four steps. Also mention the temperature and reaction conditions.

40. $\text{C}_5\text{H}_{13}\text{N} \xrightarrow[\text{N}_2]{\text{NaNO}_2/\text{HCl}} \text{Y}$
optically active 'X' (1995)

(tertiary alcohol) + other products

Find X and Y. Is Y optically active ? Write the intermediate steps. (2005)

ANSWER KEY

EXERCISE - 1 : (Basic Objective Questions)

- | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (b) | 2. (d) | 3. (c) | 4. (a) | 5. (d) | 6. (c) | 7. (d) | 8. (a) |
| 9. (b) | 10. (c) | 11. (b) | 12. (b) | 13. (b) | 14. (b) | 15. (a) | 16. (d) |
| 17. (d) | 18. (d) | 19. (a) | 20. (a) | 21. (d) | 22. (c) | 23. (d) | 24. (c) |
| 25. (d) | 26. (b) | 27. (b) | 28. (b) | 29. (d) | 30. (a) | 31. (c) | 32. (c) |
| 33. (b) | 34. (a) | 35. (d) | 36. (a) | 37. (d) | 38. (a) | 39. (c) | 40. (b) |
| 41. (d) | 42. (b) | 43. (c) | 44. (a) | 45. (b) | 46. (c) | 47. (d) | 48. (c) |
| 49. (c) | 50. (b) | 51. (b) | 52. (d) | 53. (c) | 54. (c) | 55. (a) | 56. (d) |
| 57. (d) | 58. (b) | 59. (b) | 60. (a) | 61. (a) | 62. (c) | 63. (c) | 64. (b) |
| 65. (c) | | | | | | | |

EXERCISE - 2 : (Previous Year JEE Mains Questions)

- | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (d) | 2. (c) | 3. (c) | 4. (d) | 5. (b) | 6. (c) | 7. (d) | 8. (b) |
| 9. (b) | 10. (d) | 11. (c) | 12. (a) | 13. (c) | 14. (b) | 15. (c) | 16. (a) |
| 17. (d) | | | | | | | |

JEE Mains Online

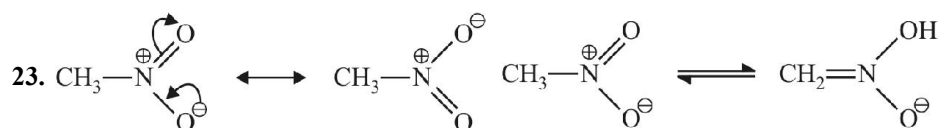
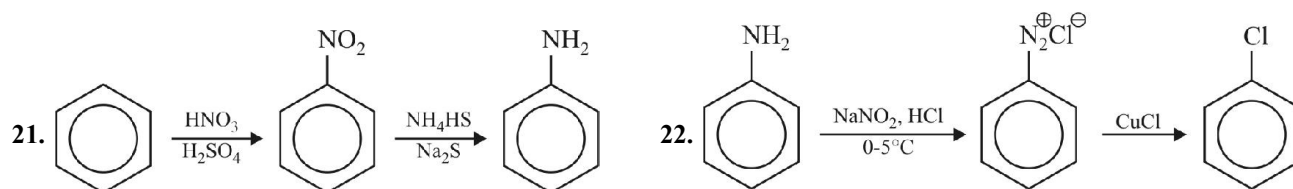
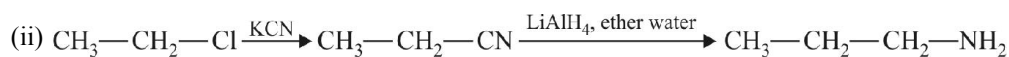
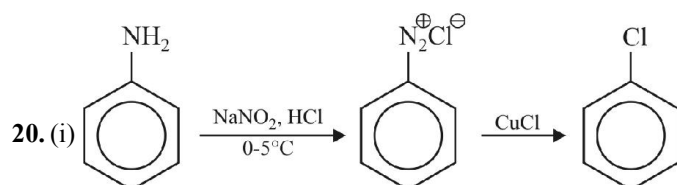
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|--------|---------|--------|--------|--------|--------|--------|--------|
| 1. (b) | 2. (d) | 3. (c) | 4. (c) | 5. (d) | 6. (b) | 7. (d) | 8. (c) |
| 9. (b) | 10. (a) | | | | | | |

EXERCISE - 3 : (Advanced Objective Questions)

- | | | | | | | | |
|------------|-----------|---------------|------------|----------------------------------|-----------|-----------|-----------|
| 1. (b) | 2. (d) | 3. (d) | 4. (c) | 5. (c) | 6. (b) | 7. (c) | 8. (c) |
| 9. (d) | 10. (d) | 11. (a, c, d) | 12. (a) | 13. (b) | 14. (c) | 15. (c) | 16. (bcd) |
| 17. (c, d) | 18. (acd) | 19. (bd) | 20. (abcd) | 21. (c) | 22. (d) | 23. (a) | 24. (c) |
| 25. (a) | 26. (d) | 27. (b) | 28. (b) | 29. (a) | 30. (abc) | 31. (abc) | 32. (a) |
| 33. (d) | 34. (c) | 35. (c) | 36. (a) | 37. (A-p,s; B-p,s; C-r; D-p,q,s) | | | 38. (ac) |
| 39. (d) | 40. (c) | 41. (a) | 42. (d) | 43. (b) | 44. (ac) | 45. (abc) | 46. (bcd) |
| 47. (a) | 48. (b) | 49. (c) | 50. (a) | 51. (a) | 52. (d) | 53. (d) | 54. (d) |
| 55. (a) | 56. (d) | 57. (a) | 58. (b) | 59. (a) | 60. (b) | 61. (b) | |

EXERCISE - 4 : (Previous Year JEE Advanced Questions)

1. (c) 2. (c) 3. (b) 4. (c) 5. (c) 6. (d) 7. (b) 8. (a)
9. (b) 10. (c) 11. (b) 12. (a, d) 13. (b, c, d) 14. (b, d) 15. (a, b, d) 16. (e)
17. (a) 18. o-nitrophenol 19. Zwitter ion



Dream on !!

